I 

Pl ant s

# This week

**Thu Oct 29**

**NO CLASS**
**midterm work day**

**Thu 3:10 PM**

**group survey (Canvas)**

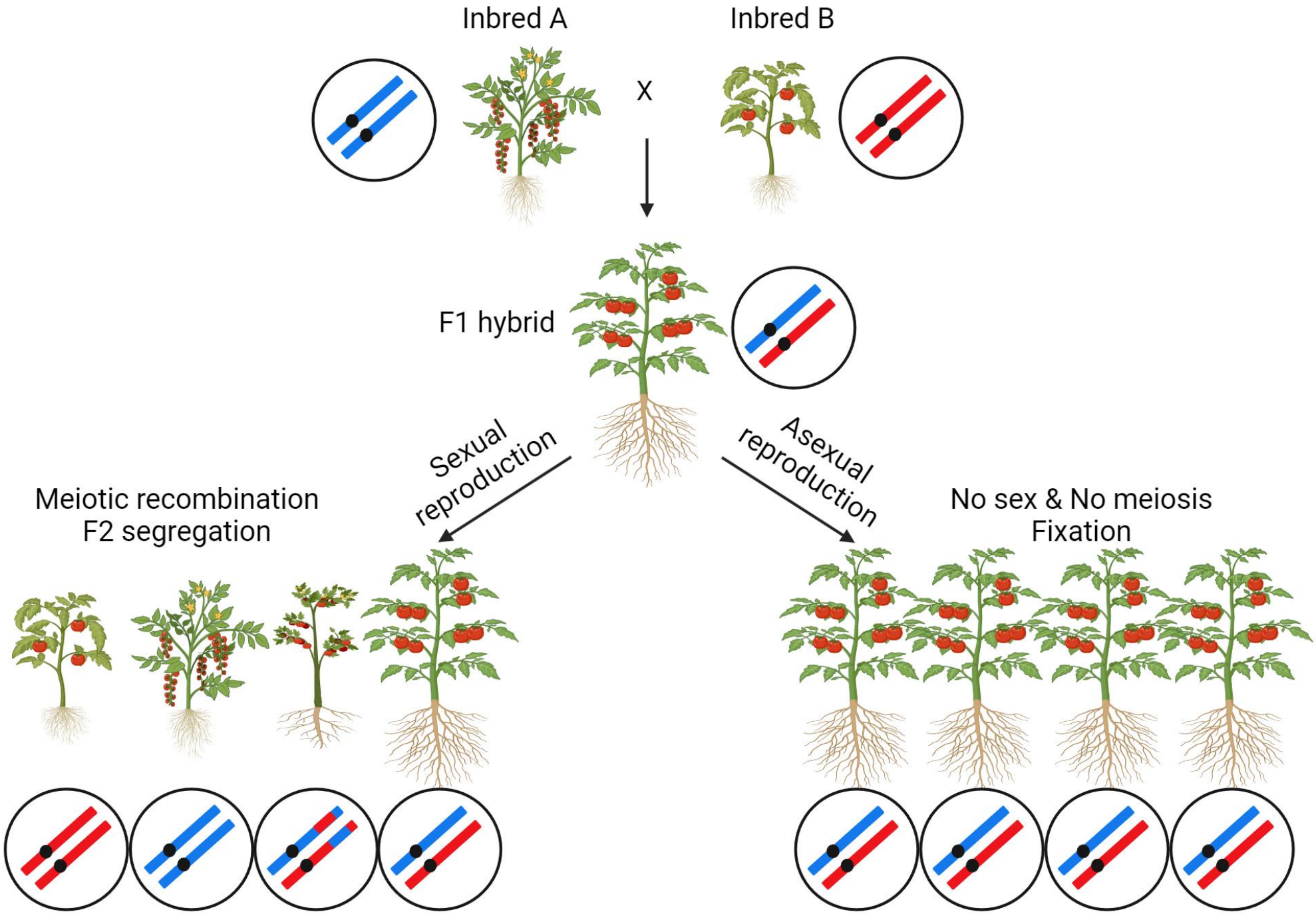
**Tue Nov 3**

**Quiz 5**
**= today's lecture**

**Tue Nov 10**

**midterm due 3:10 PM**

**Today = Tuesday + Thursday's material**


Inbred A
Inbred B
X
F1 hybrid
Sexual reproduction
Asexual reproduction
Meiotic recombination
F2 segregation
No sex & No meiosis
Fixation

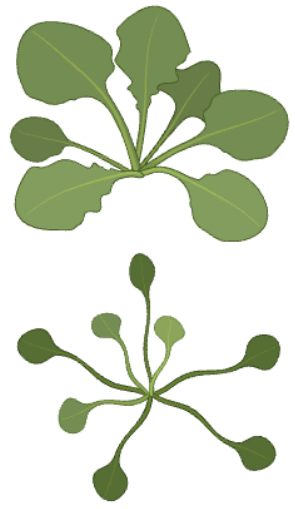
## “Forward” genetics

What gene underlies a phenotype?
(**Ph**)orward…genetics

Often used together

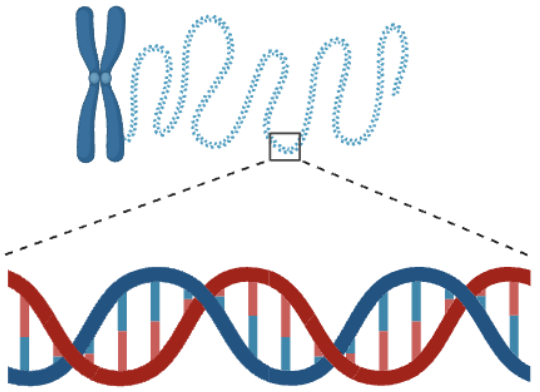
## “Reverse” genetics

What phenotype does a gene effect?

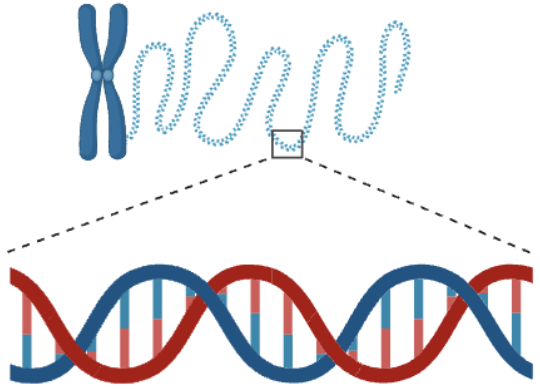


PHENOTYPE

→ ?

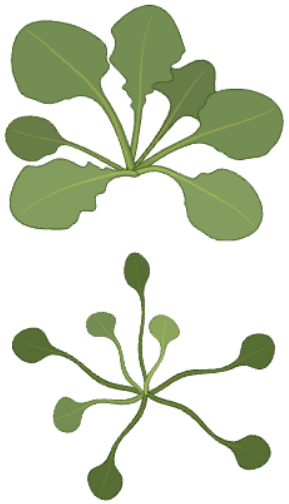


GENE



GENE

→ ?



PHENOTYPE

# Phenotype → gene: four routes

**Mutant screen**

break genes, find the weird plant

**MutMap**

pool mutants, sequence

**QTL mapping**

cross two lines

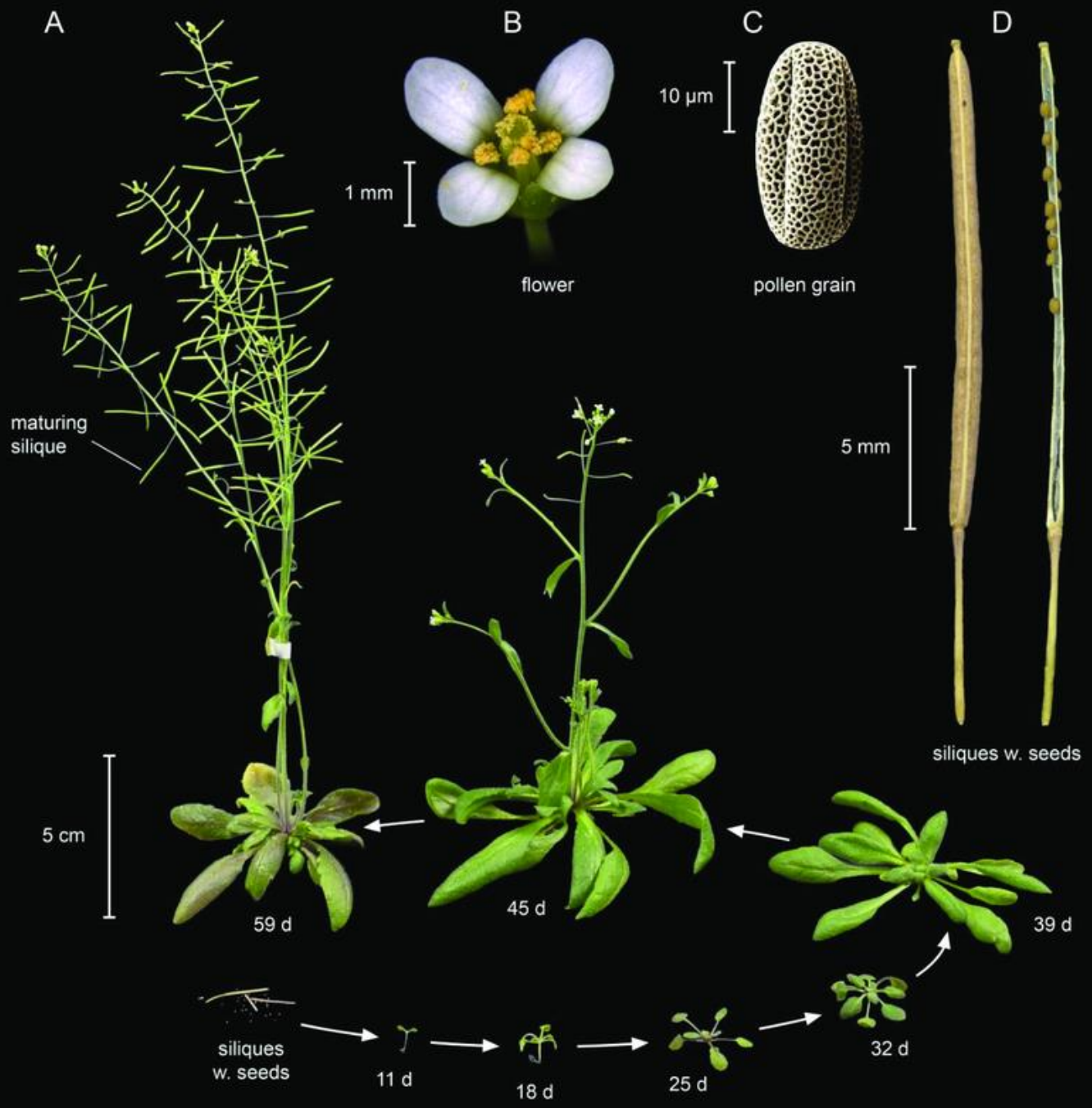
**GWAS**

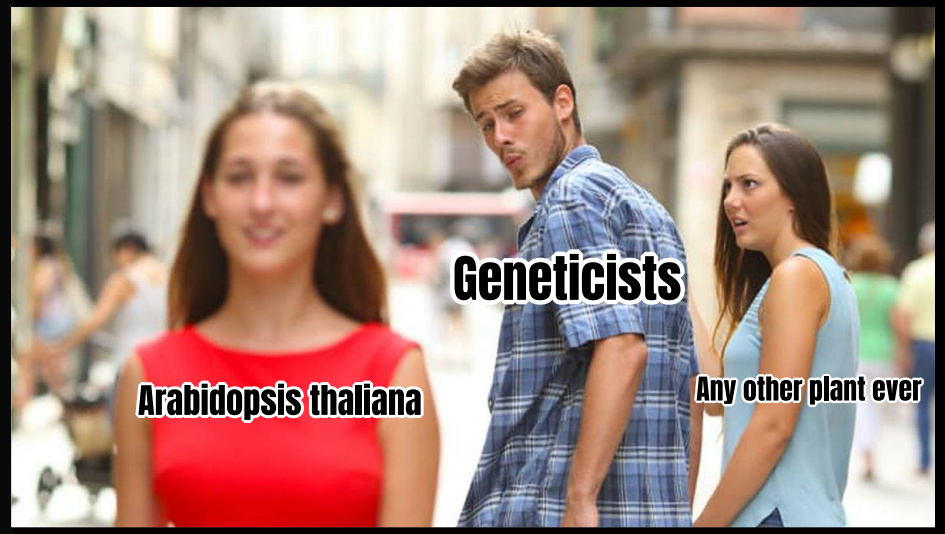
natural population

make new mutations

use natural variation

**Then: is it really that gene? (complementation)**


A
B
C
D
1 mm
flower
10 µm
pollen grain
5 mm
siliques w. seeds
maturing
silique
5 cm
59 d
45 d
39 d
siliques
w. seeds
11 d
18 d
25 d
32 d


Arabidopsis thaliana
Geneticists
Any other plant ever


PLANT BREEDERS
ARABIDOPSIS


Bernie
Plant
Scientists
I am once again asking
To conduct a study on arabidopsis thaliana
imgflip.com

# Live poll

Text wall

## What could you NOT study with Arabidopsis?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-fg-arabidopsis

host screen

# Why this little weed?

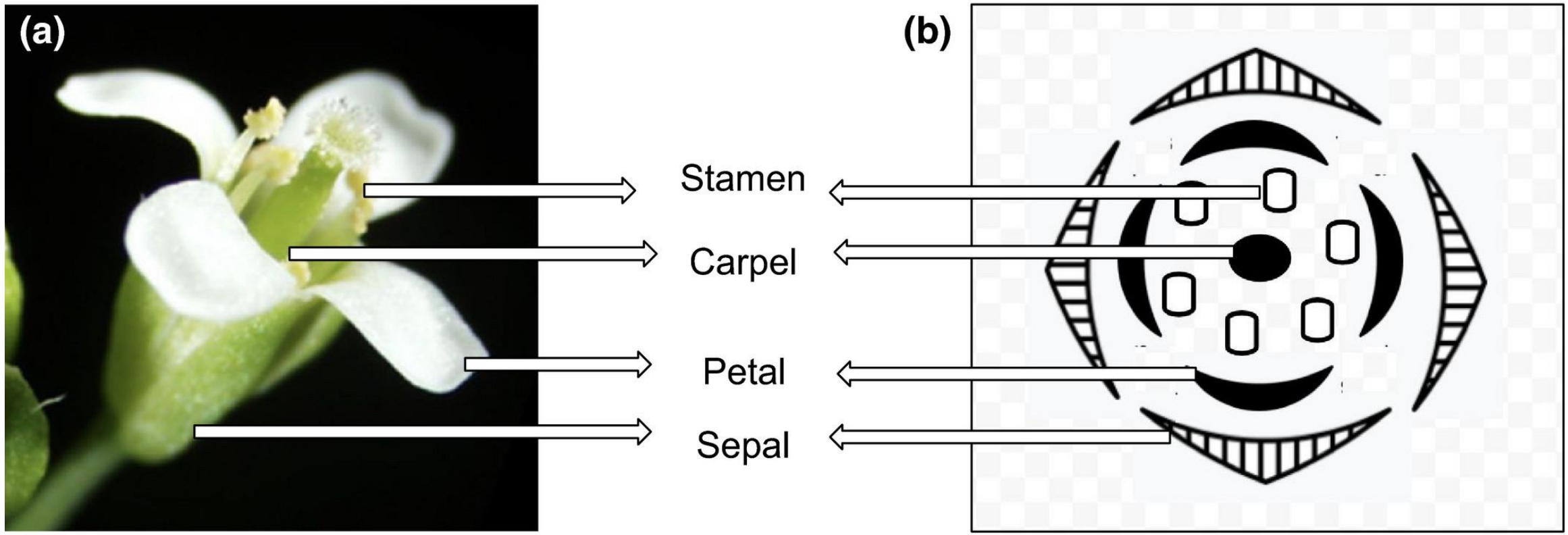
~135 Mb
small genome

~6 weeks
seed to seed

selfs
but easy to
cross

tiny
hundreds per
tray

**NOT a crop**


(a)
(b)
Stamen
Carpel
Petal
Sepal



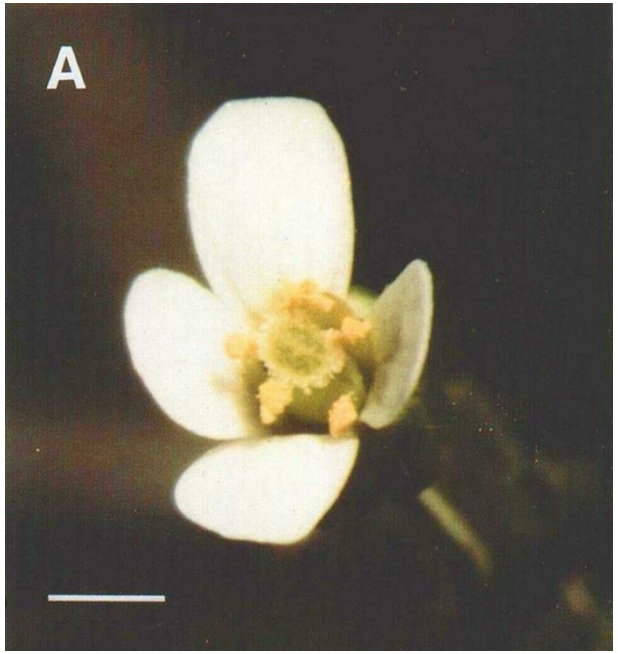




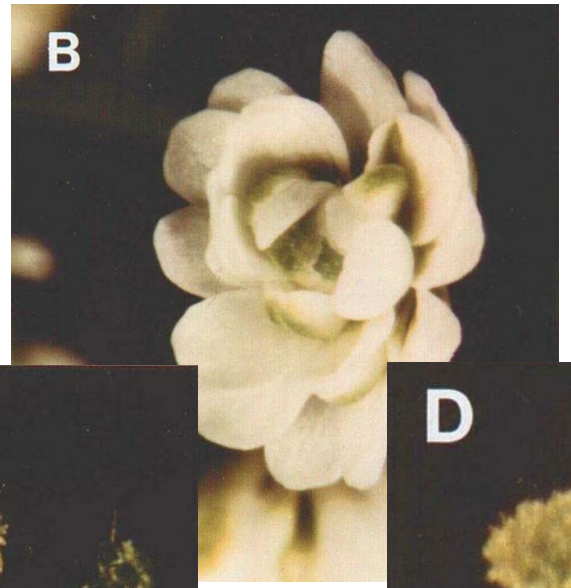


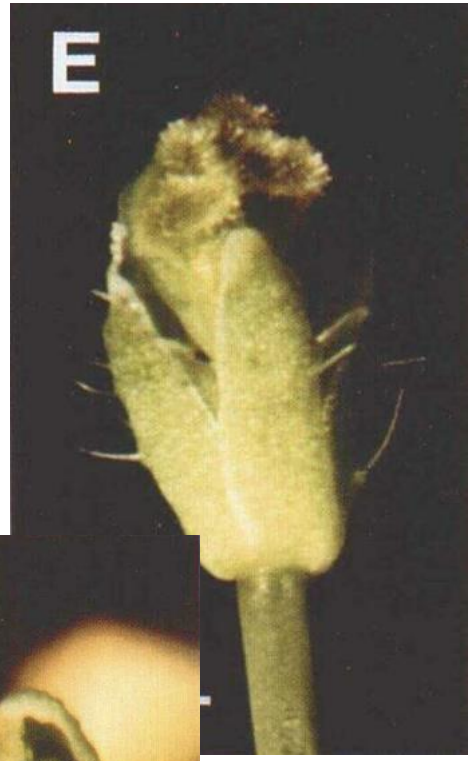

A
ap1-1
B
ap3-3
C
ag-1
D
WT

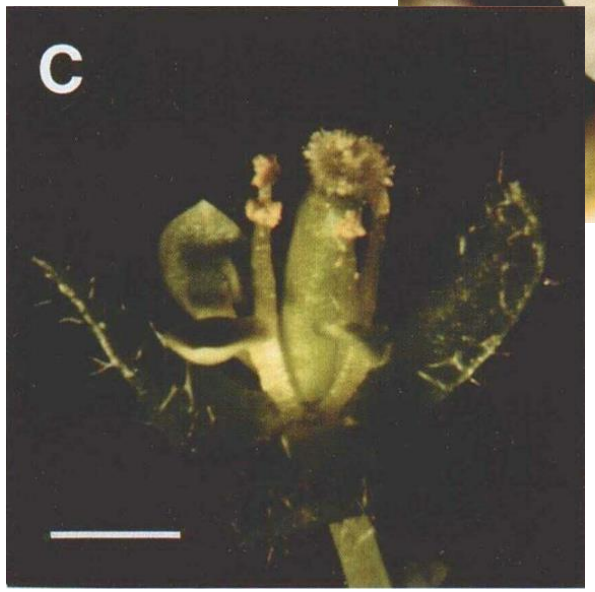
(A) Wild-type.
(B) *agamous*.
(C) *apetala2*.
(D) *apetala3*.
(E) *pistillata*.




+EMS mutagen



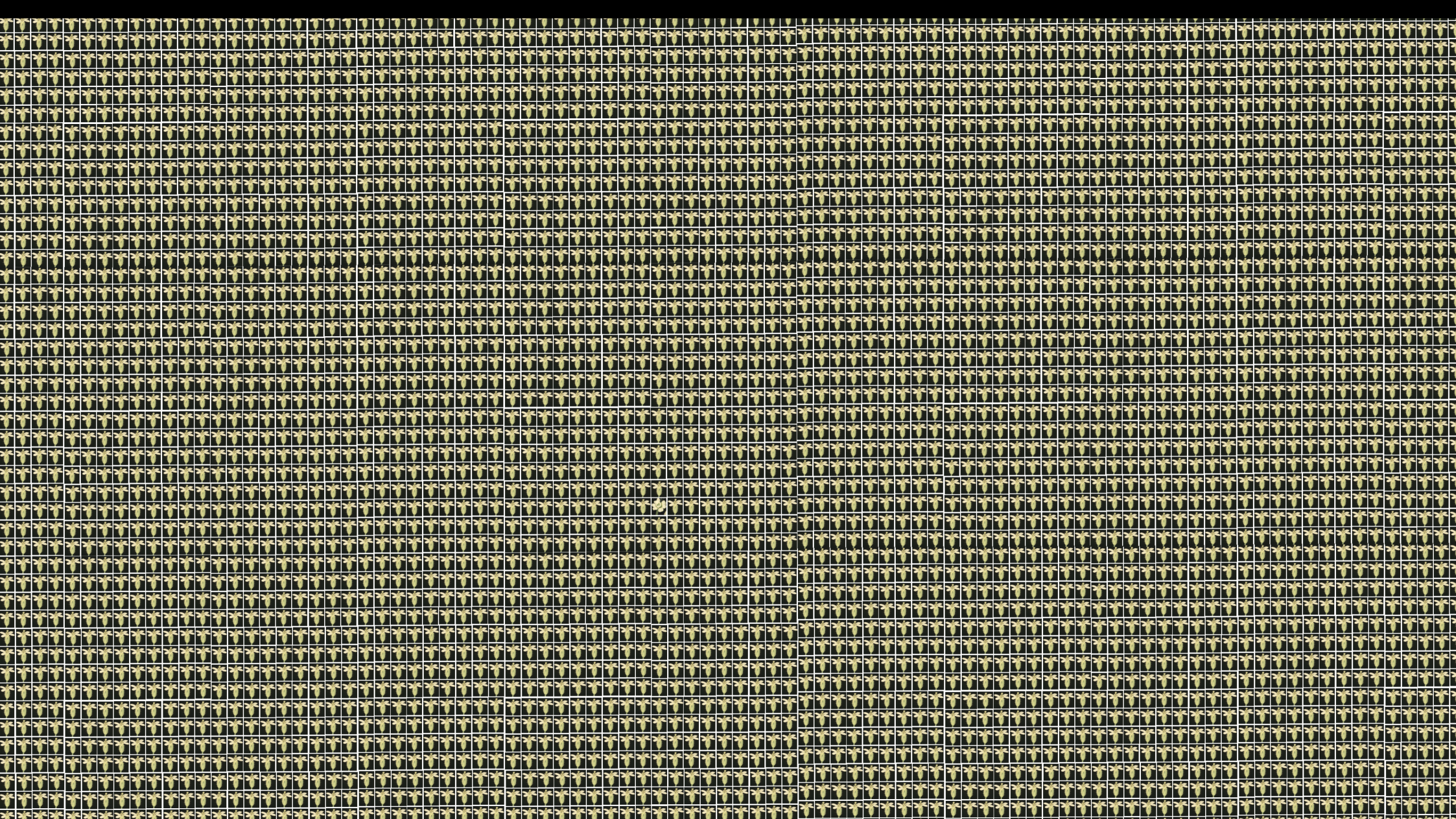


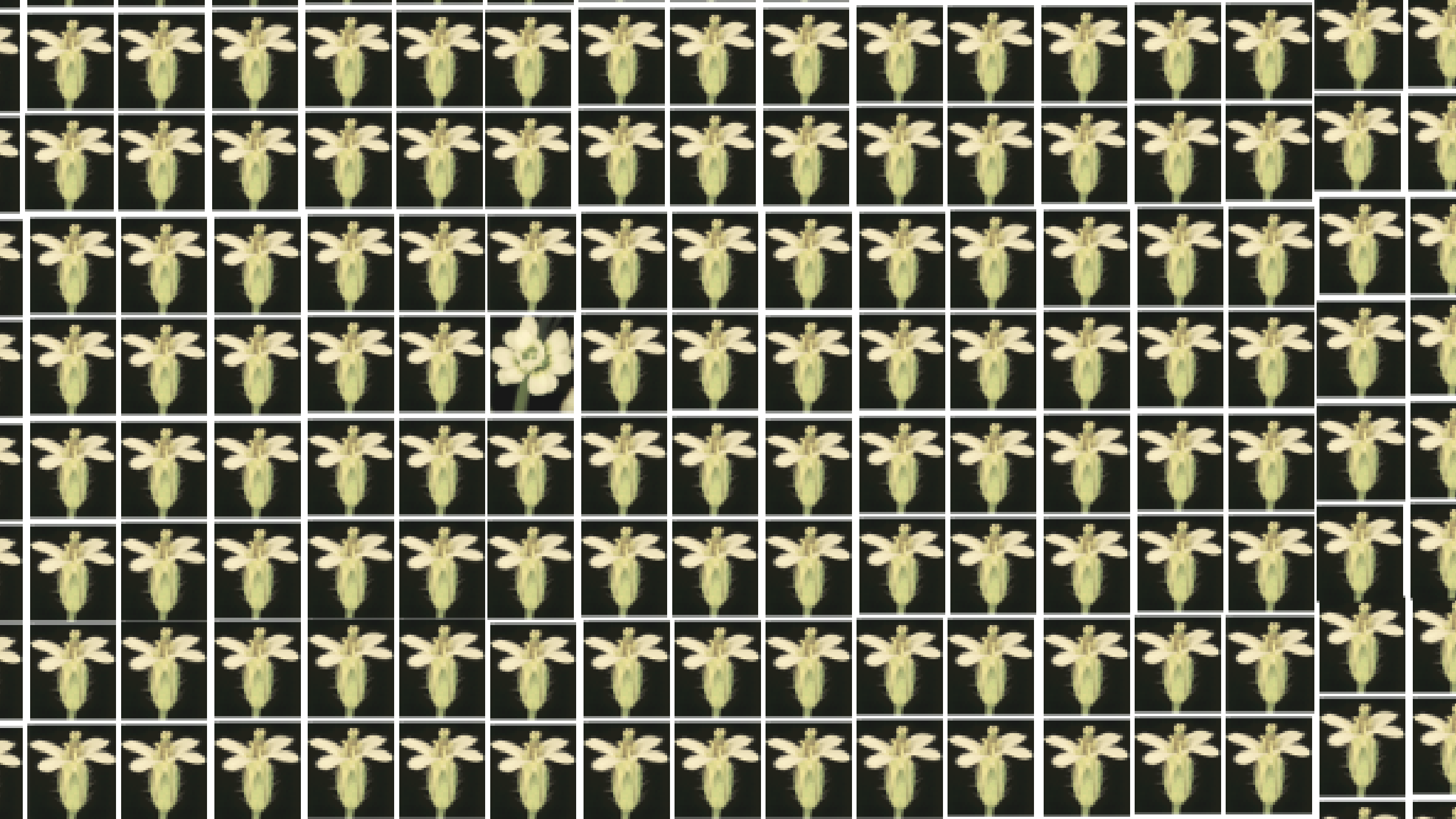


**What are the genes?**

# MUTANT SCREEN

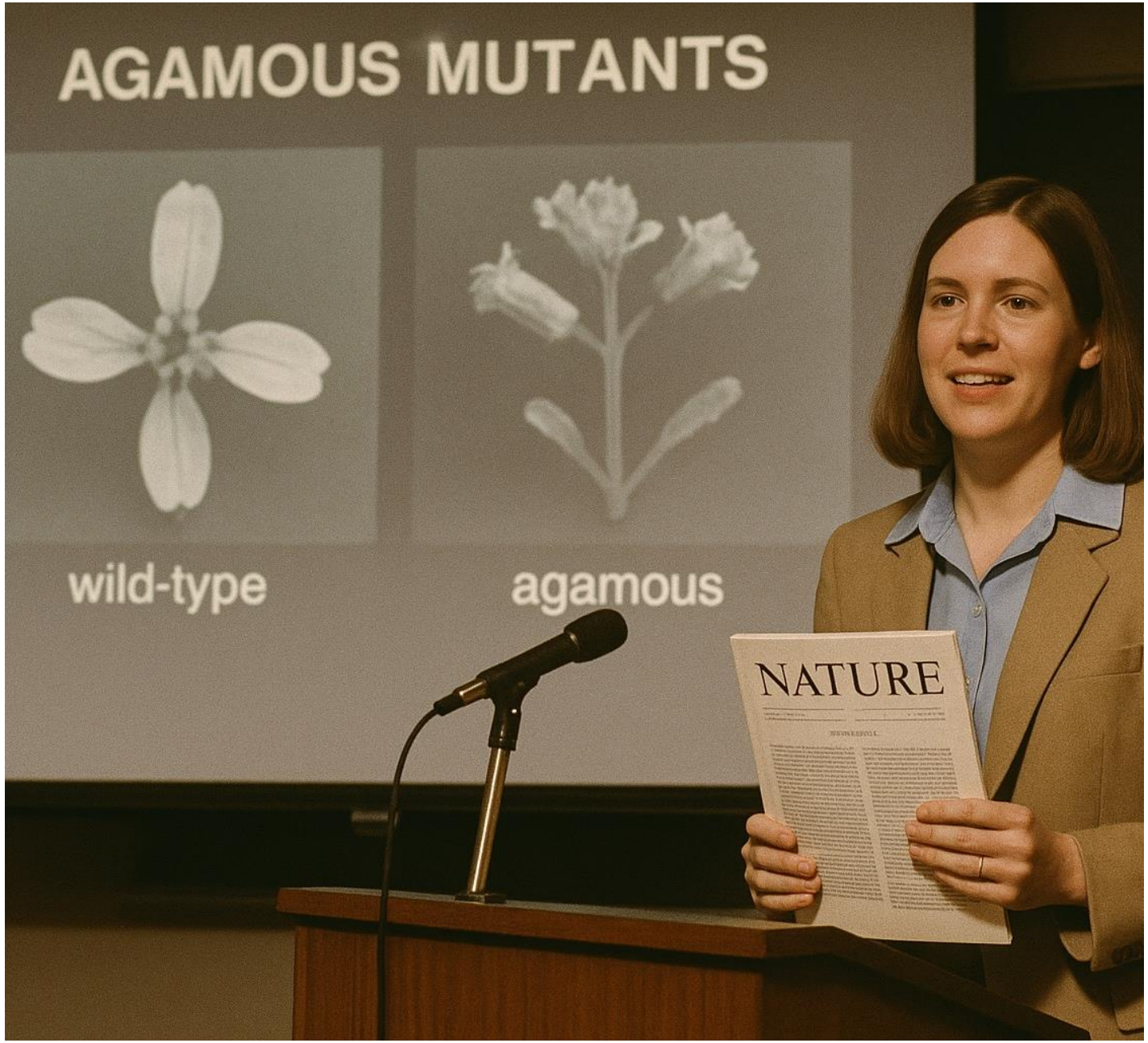

MUTANT SCREEN

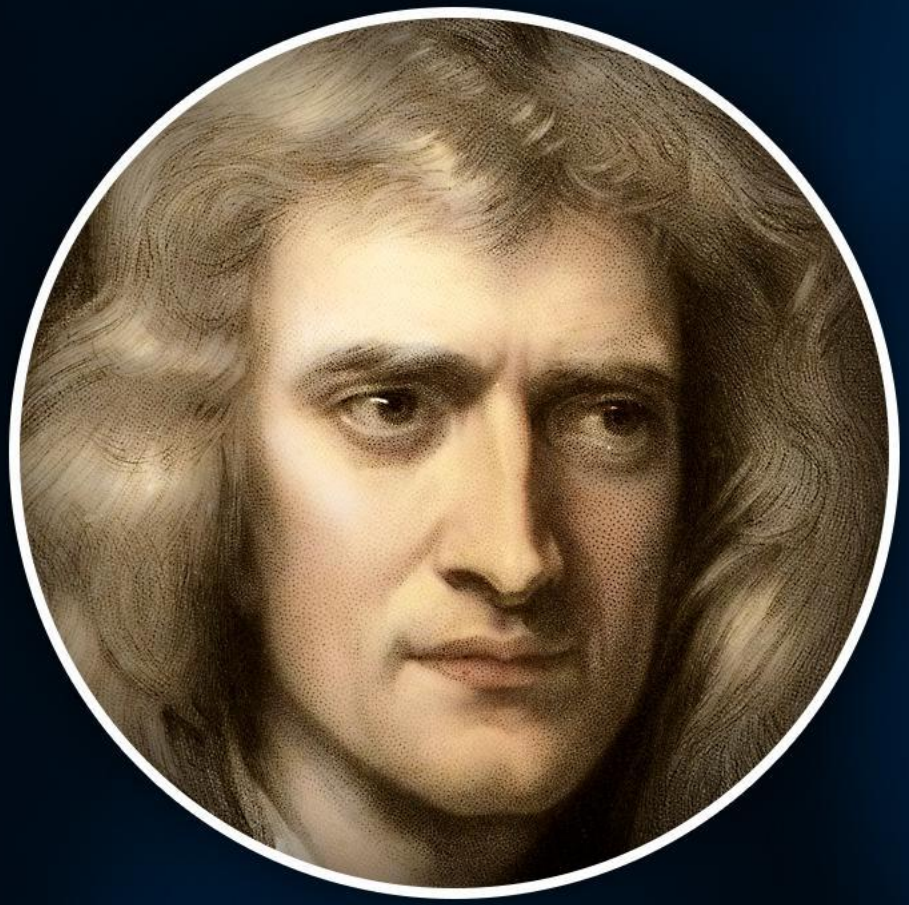





KpnI
Sau3AI
MseI
RsaI
SspI
TaqI
MW
MW
AluI
BamHI
BfaI
BstUI
DraI
EcoRI
CauPh-YNym
B


DISSERTATION
BY


AGAMOUS MUTANTS
wild-type
agamous
NATURE

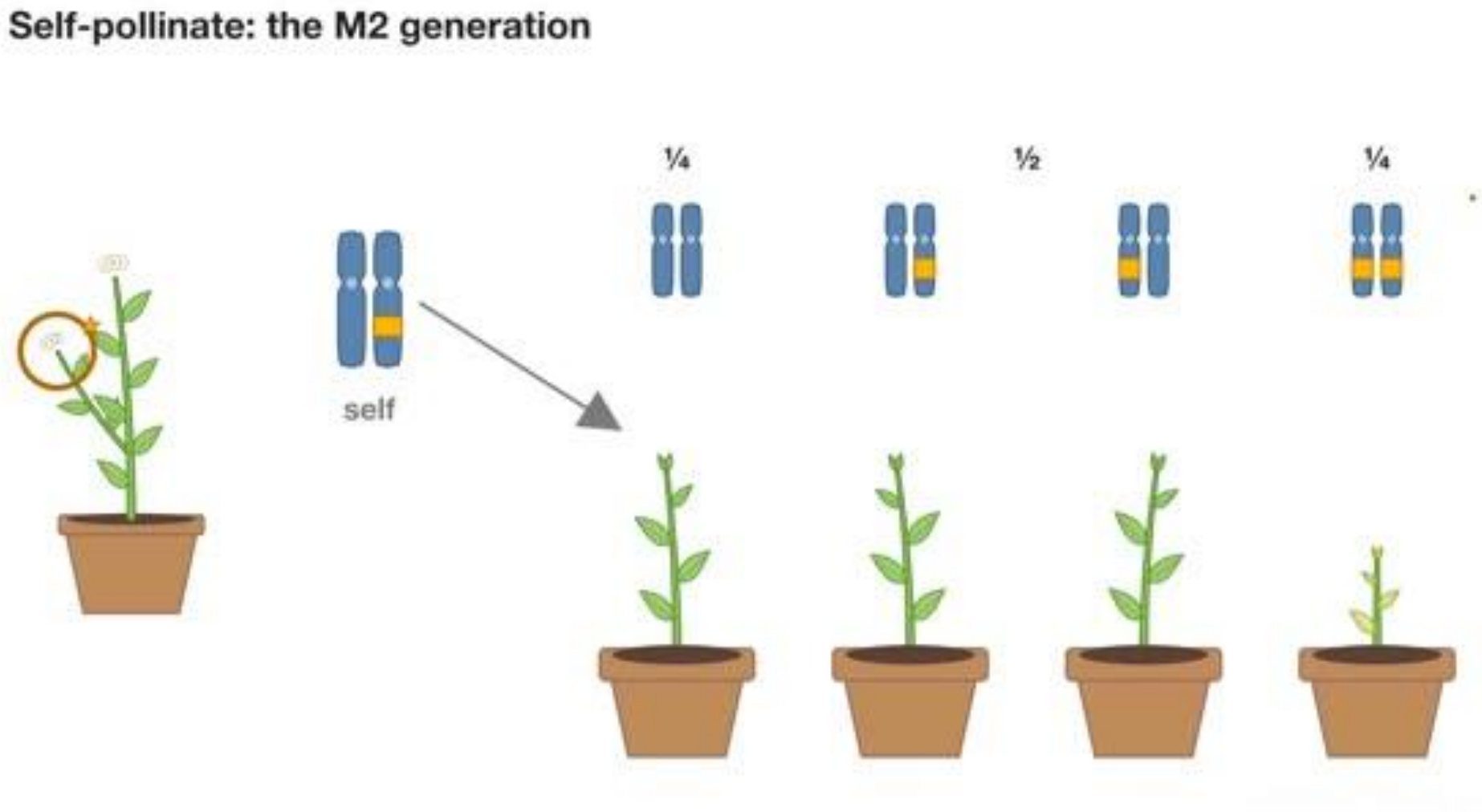


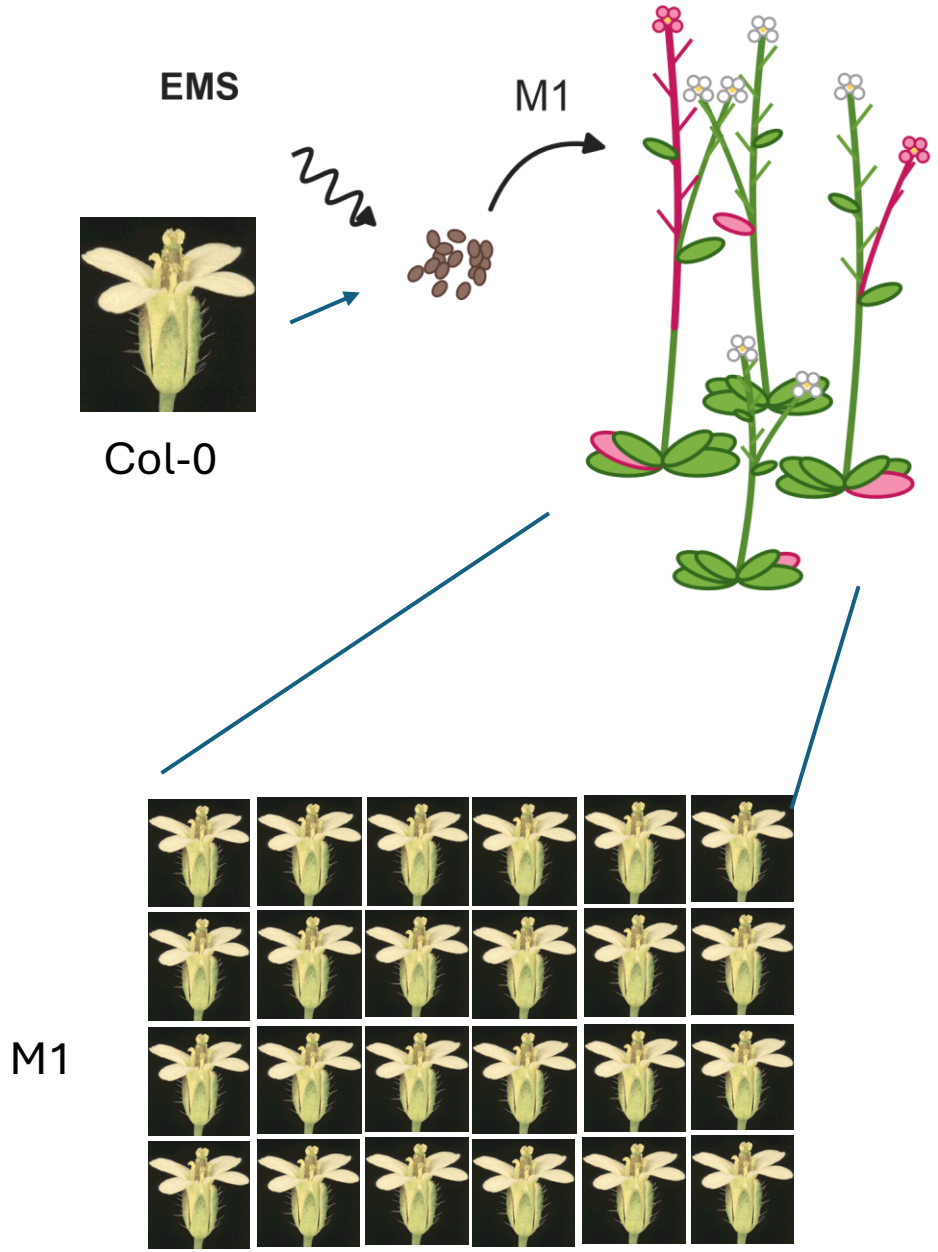
If I have seen further, it is by standing on the shoulders of giants.

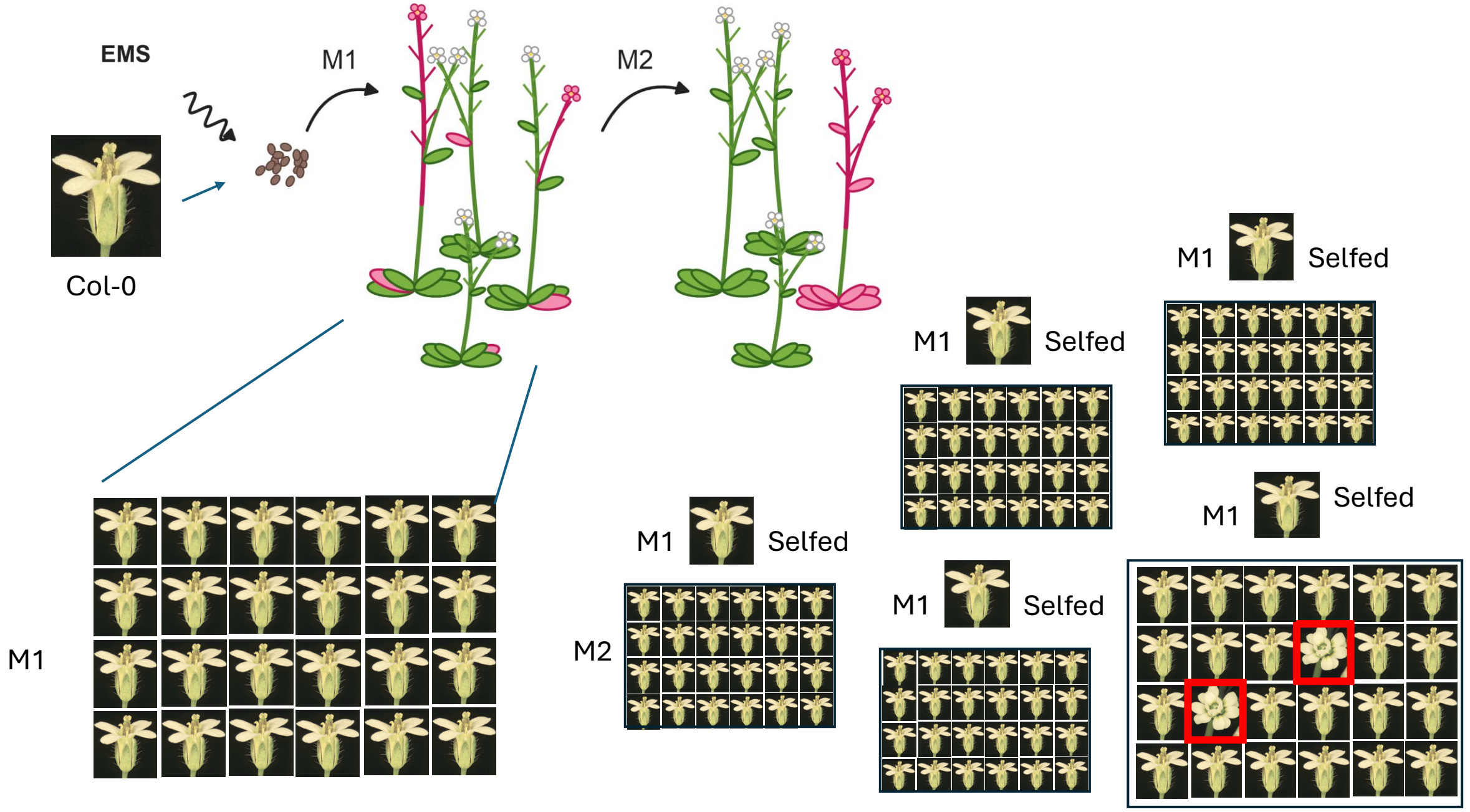
~ Isaac Newton

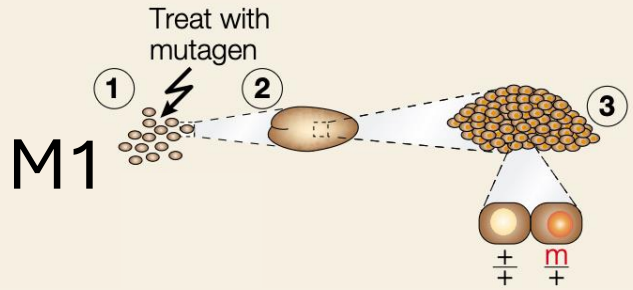
@NateTalksToYou

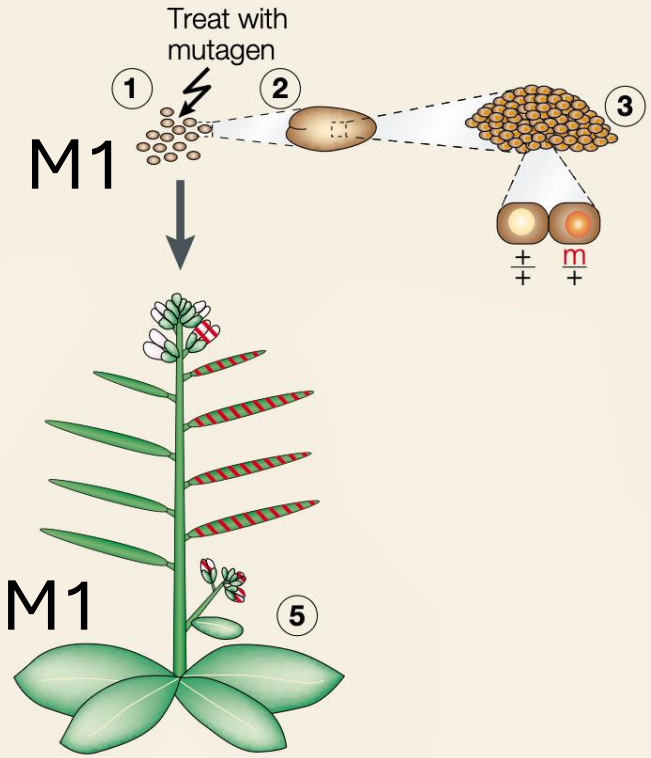
# Recap (Oct 6): why screen the M2?

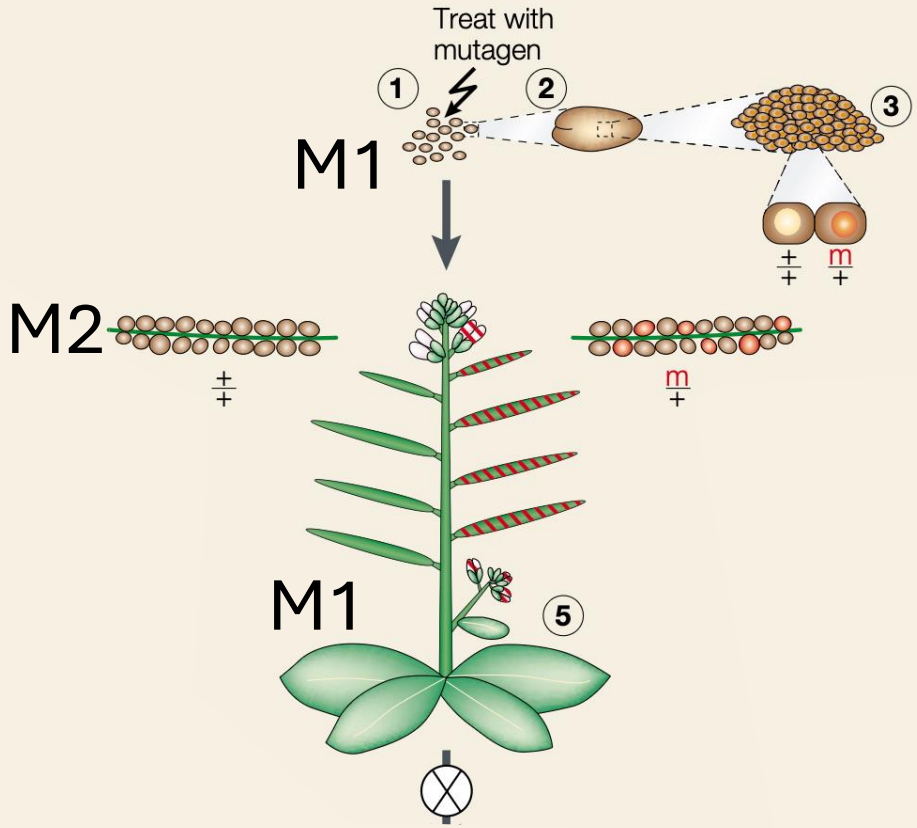


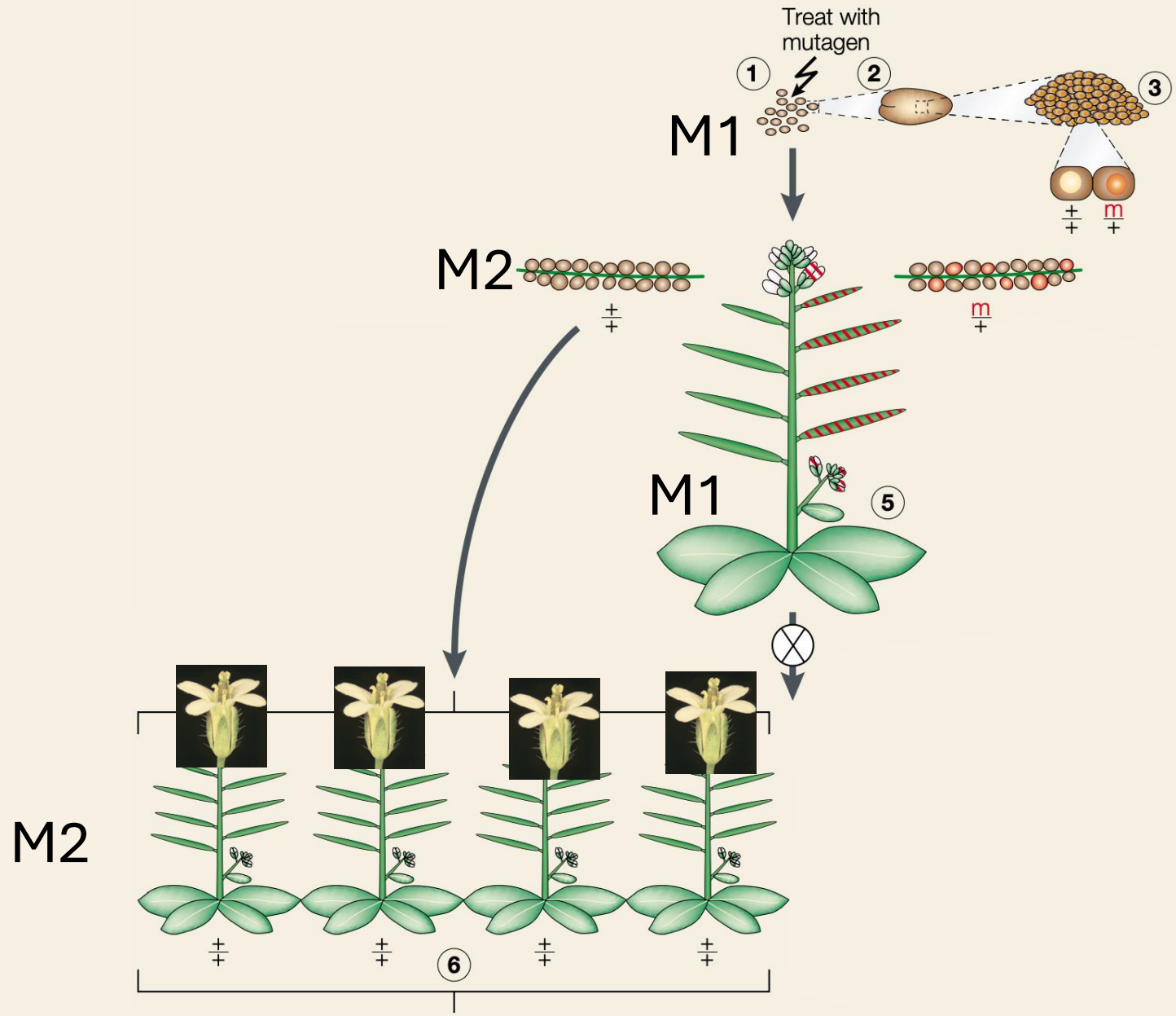

EMS
M1
Col-0
M1

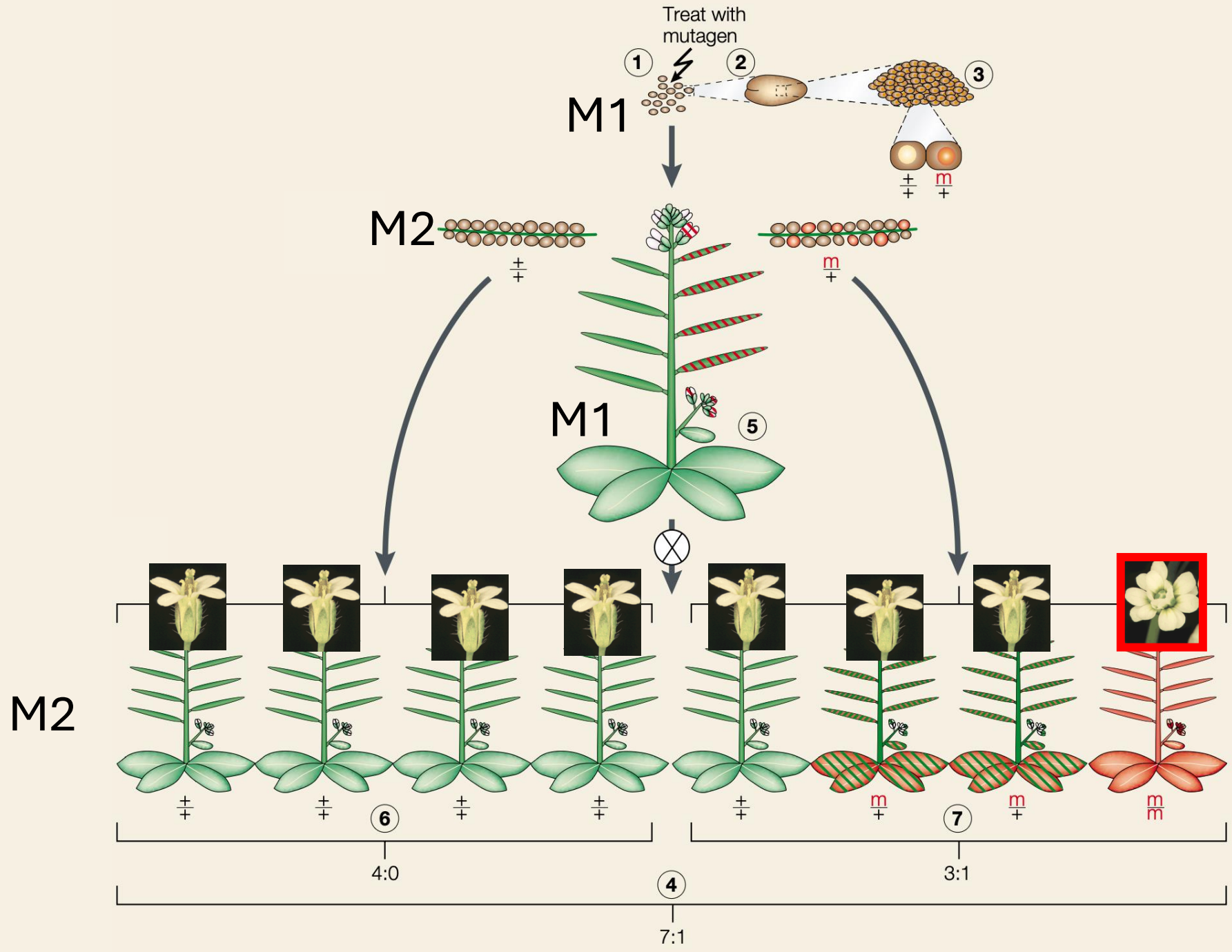

EMS
M1
M2
Col-0
M1
M1
Selfed
M2
M1
Selfed
M1
Selfed
M1
Selfed

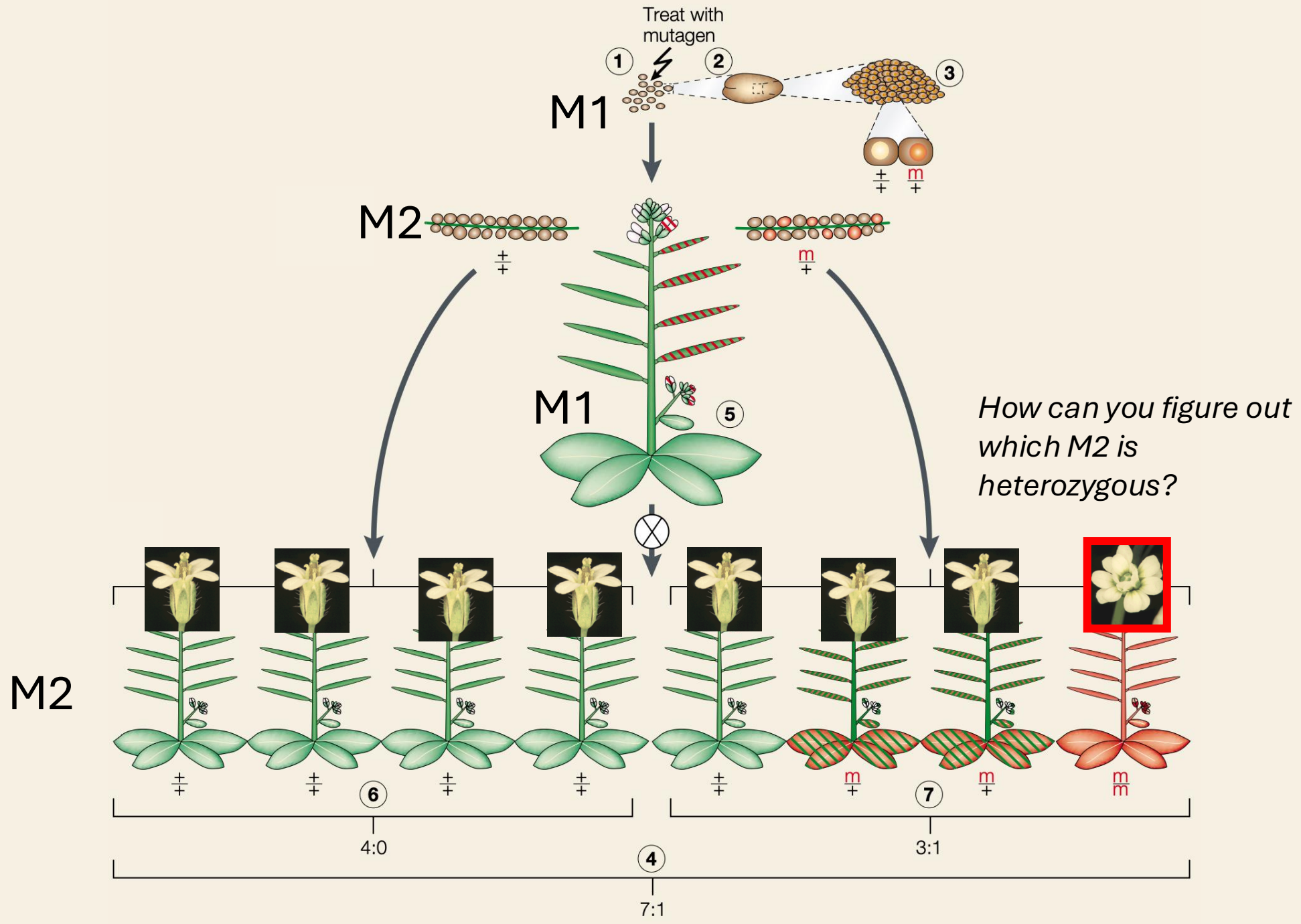

Treat with
mutagen
1
2
3
M1
+
+
m
+


Treat with
mutagen
1
2
3
M1
+
+
m
+
M1
5


Treat with mutagen
1
2
3
M1
+ +
m +
M2
+ +
m +
M1
5


Treat with mutagen
1
2
3
M1
+ +
m +
M2
+ +
m +
M1
5
M2
+ +
+ +
+ +
+ +
6


Treat with mutagen
1
2
3
M1
+/+
m/+
M2
+/+
m/+
M1
5
M2
+/+
+/+
+/+
+/+
m/+
m/+
m/m
6
7
4:0
3:1
4
7:1




*How can you figure out which M2 is heterozygous?*

M1 Selfed

M2

Col-0 Ler (different genotype)

WT/mut X WT/WT

F1 WT/mut

F2

DNA sequence

*What mutation do they all have in common?*

Chromosome:

1 2 3 4 5

C>T

C>T ..... Causes stop codon

AT4G18960.1

Evidence that this gene is involved in floral trait....

Gets name "***Agamous***" because its needed to produce gamete forming organs

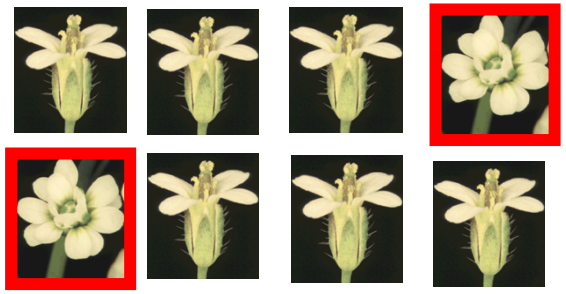
M1  Selfed

M2 

**Col-0** Ler (different genotype)

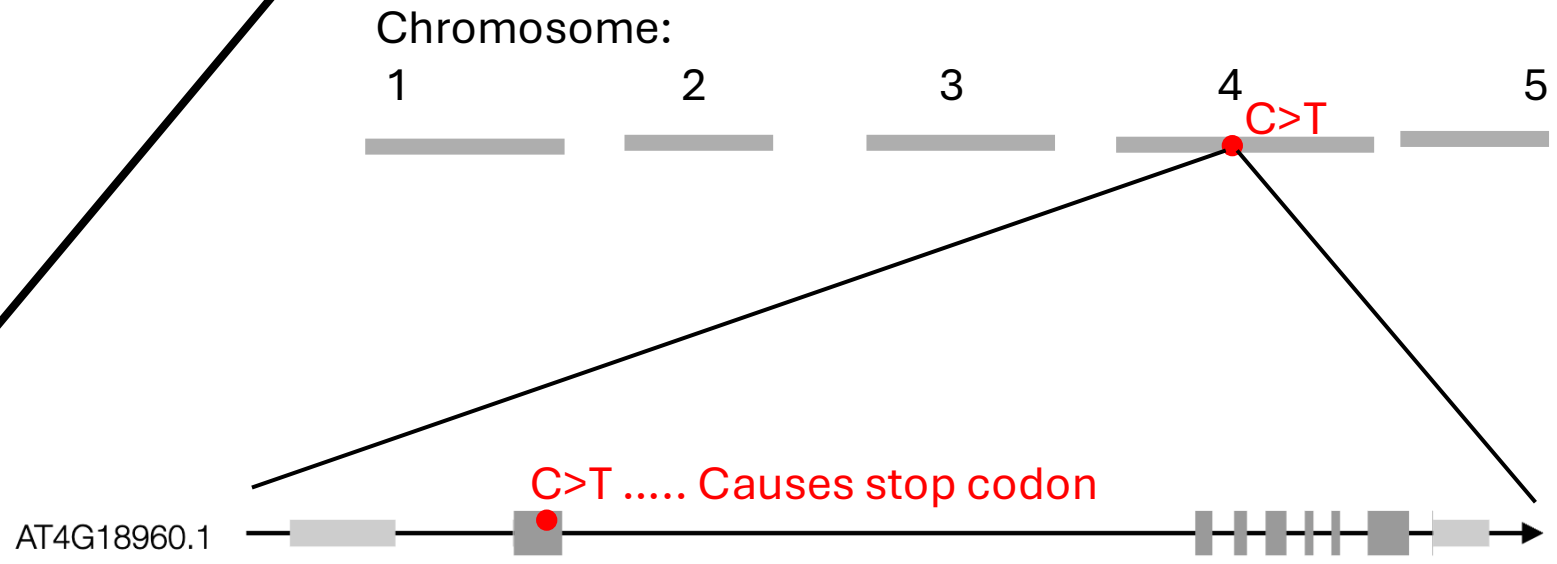
WT/mut  X  WT/WT

F1  WT/mut

F2 




*What mutation do they all have in common?*

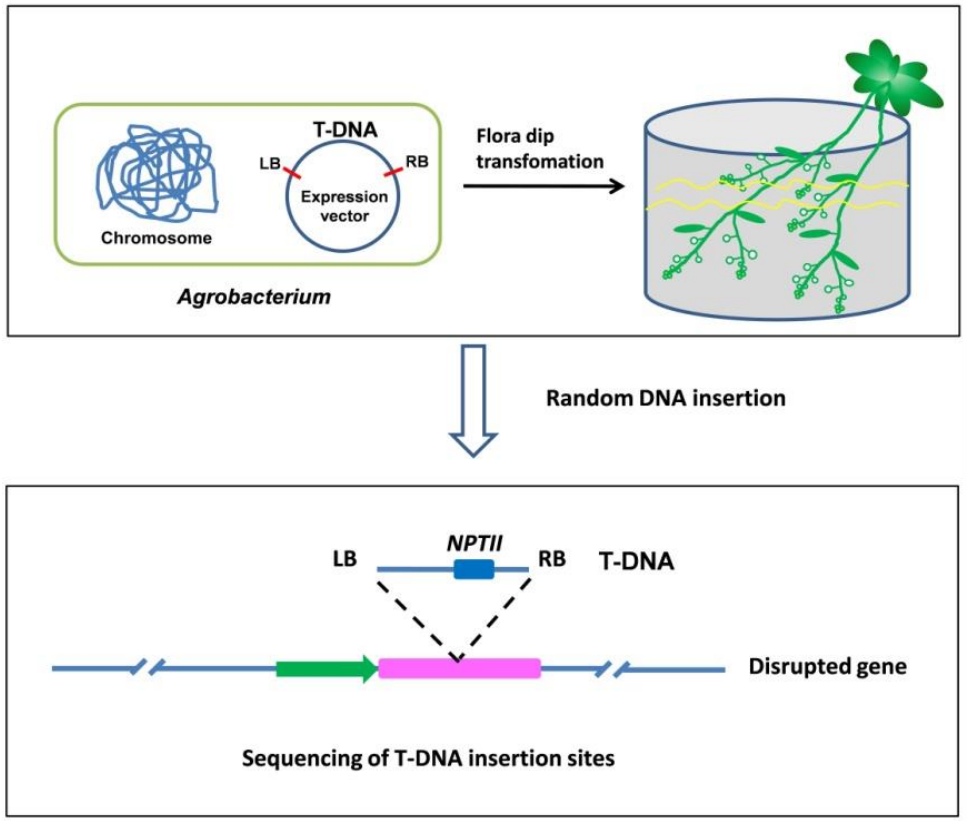



Evidence that this gene is involved in floral trait….

Gets name ”***Agamous***” because its needed to produce gamete forming organs

# Actually what they did for *agamous*

T-DNA mutagenesis

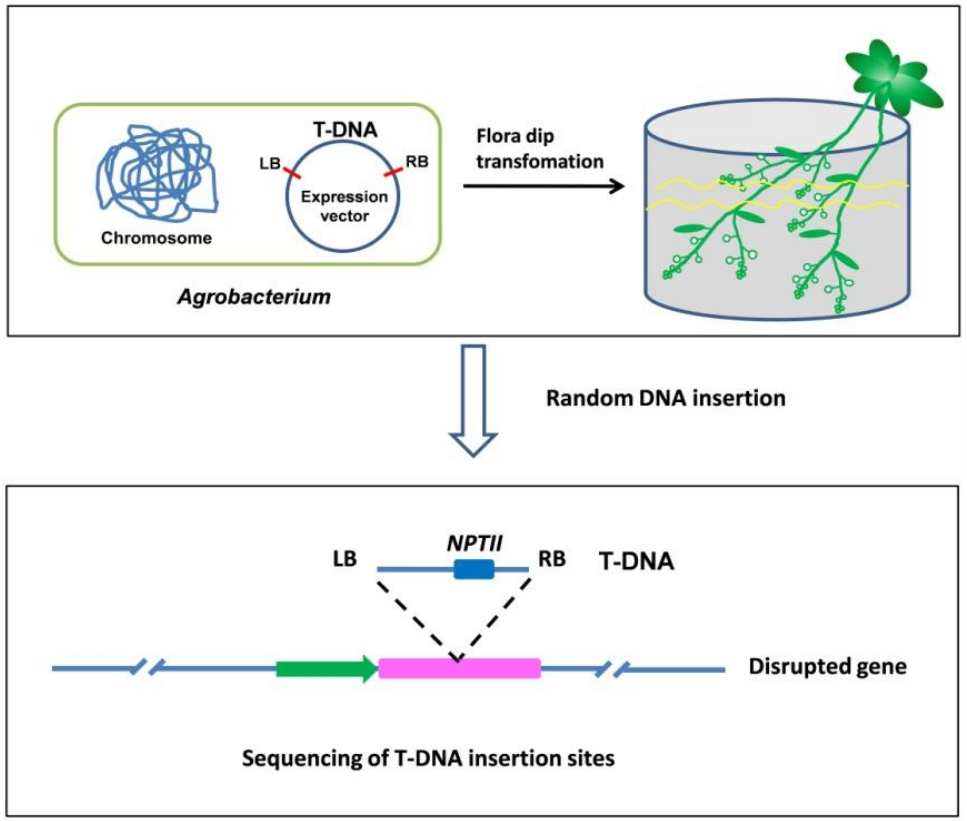



M1

**M2: look for weird phenotypes & sequence near TDNA**

# Actually what they did for *agamous*

T-DNA mutagenesis



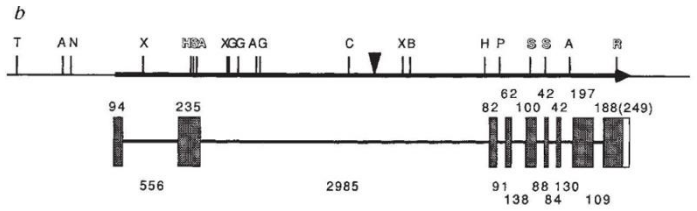

M1

**M2: look for weird phenotypes & sequence near TDNA**



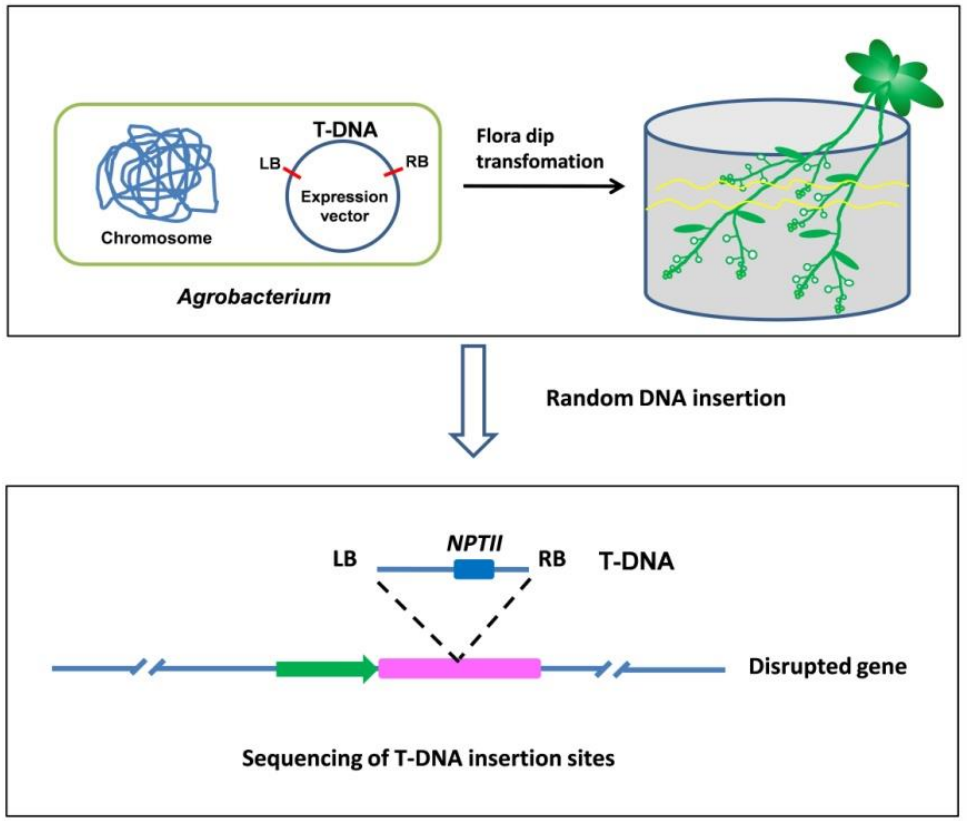
***ag-2***

Gene



# Actually what they did for *agamous*

T-DNA mutagenesis




M1

**EMS mutagenesis**



***ag-1***

**TDNA mutagenesis**



***ag-2***



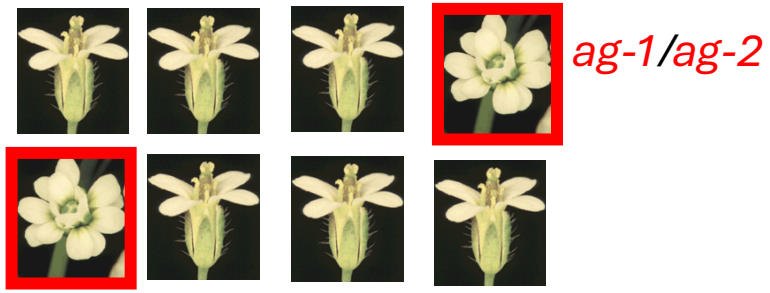
*WT/ag-1*

X



*WT/ag-2*

F1



*ag-1/ag-2*

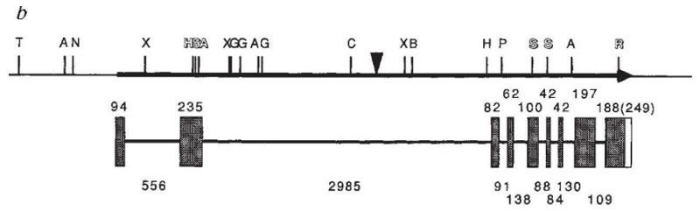
*ag-1/ag-2*

**M2: look for weird phenotypes & sequence near TDNA**



***ag-2***

Gene




**ag-1 and ag-2 are allelic**

# Complementation test



mutant 1

×

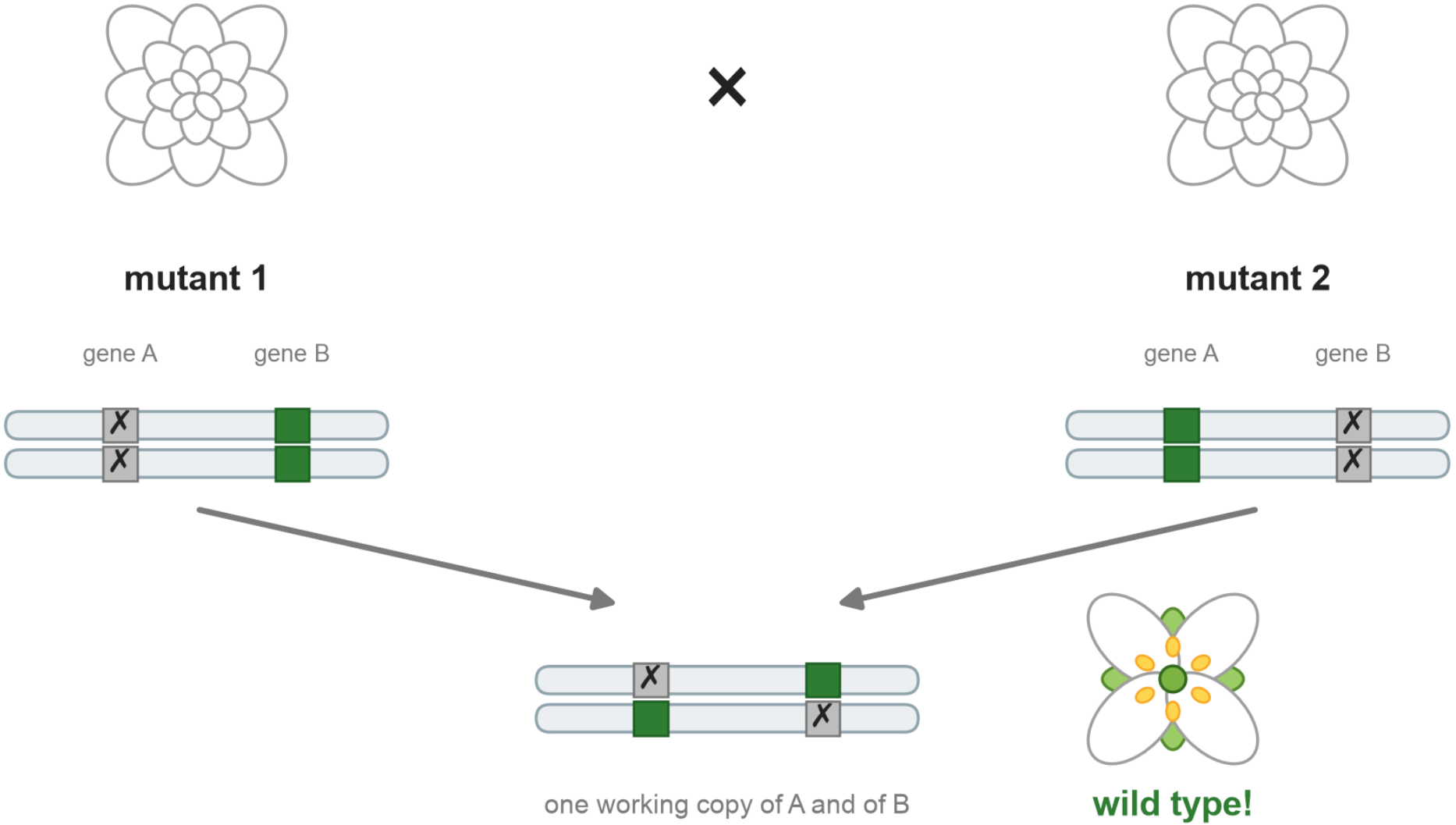


mutant 2

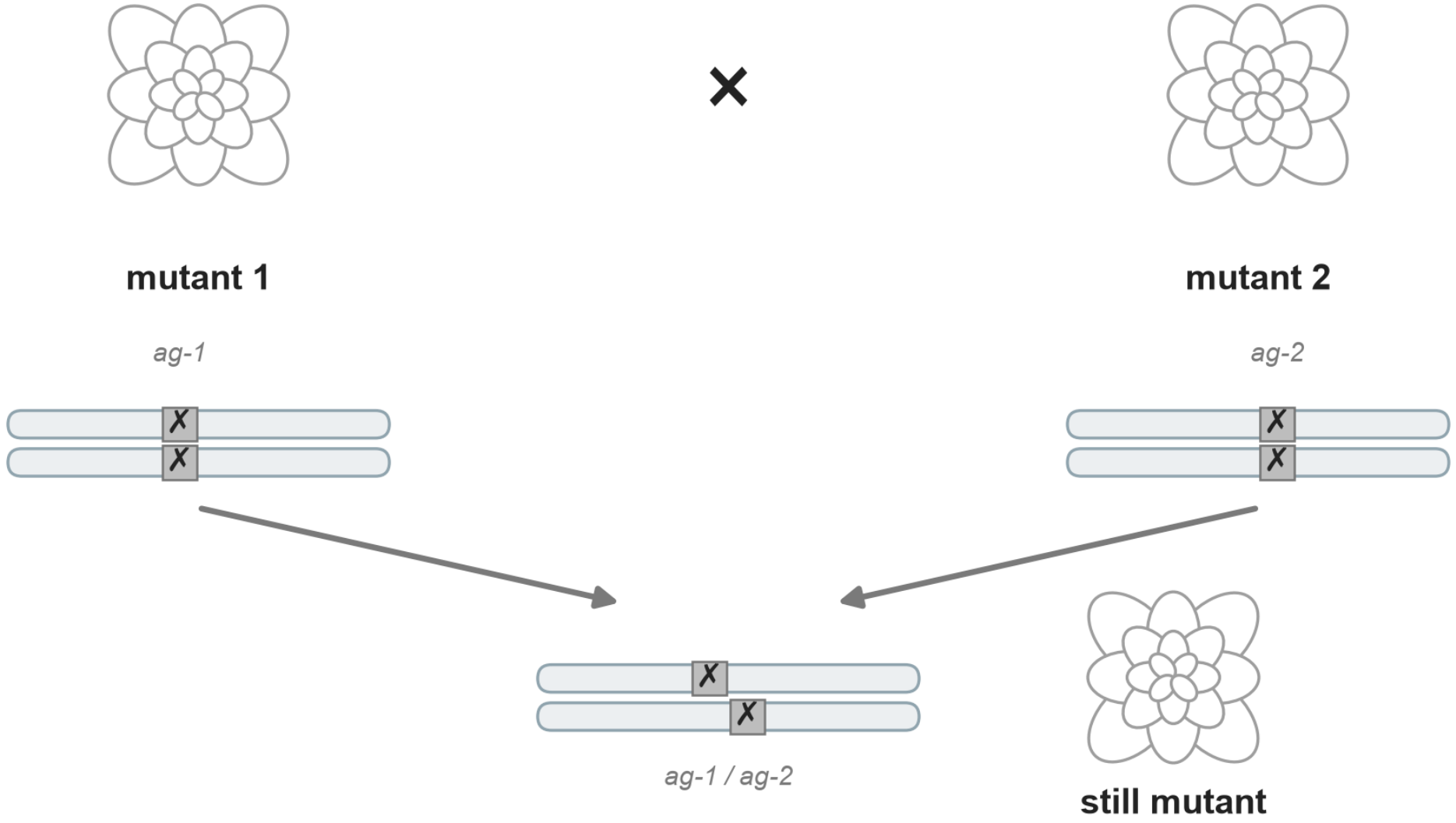
**F1 = ?**

Same broken gene, or two different broken genes?

# F1 wild type → different genes



# F1 mutant → same gene



AT4G18960.1




Gene : **AG** *A. thaliana*

| | | | |
|---|---|---|---|
| DB identifier | **AT4G18960** | Secondary Identifier | **locus:2117144** |
| Name | **AGAMOUS** | Brief Description | **K-box region and MADS-box transcription factor family protein** |
| TAIR Computational Description | **K-box region and MADS-box transcription factor family protein;(source:Araport11)** | | |

# 1991

Let's Share What We Know

World Wide Web

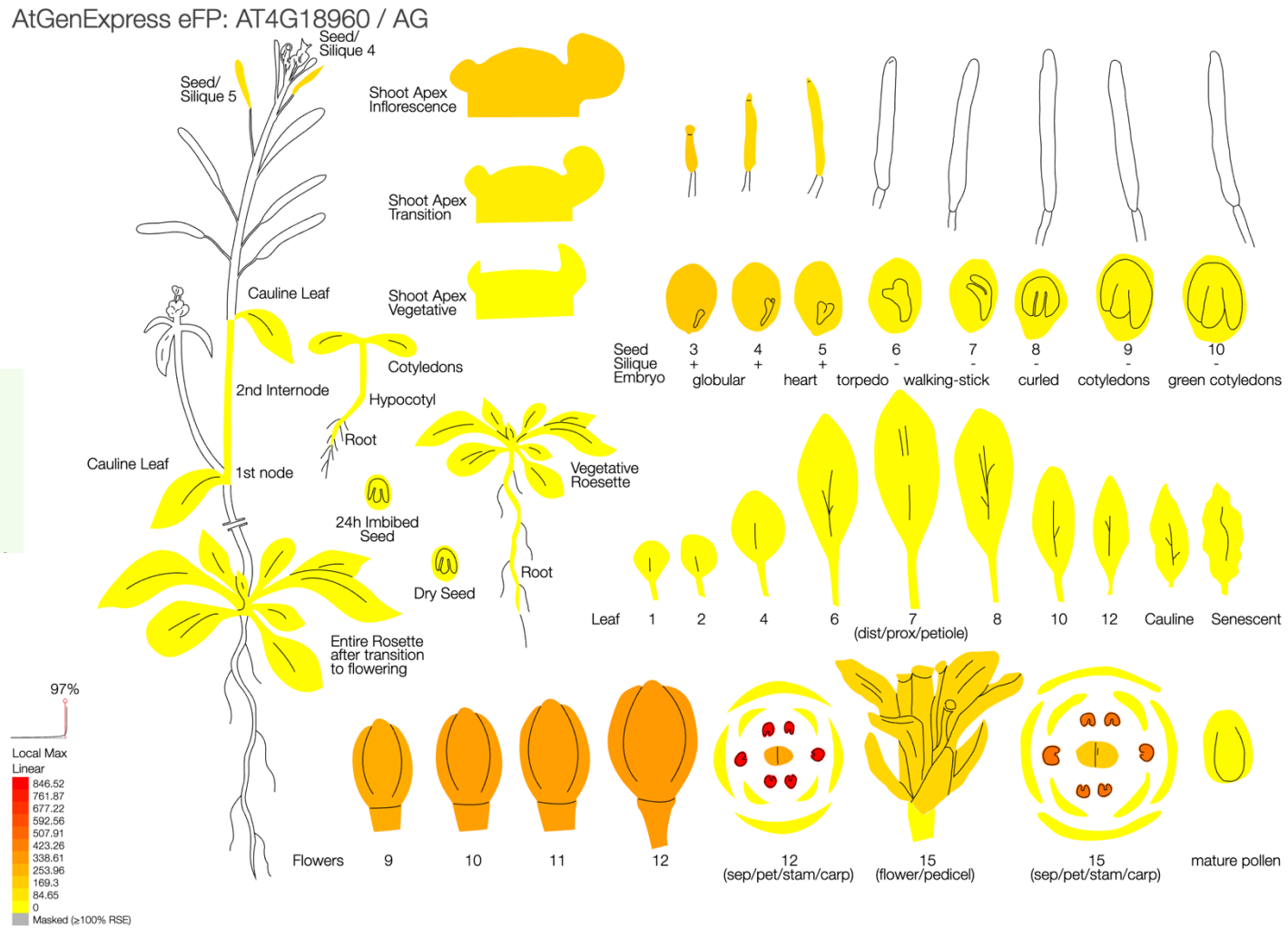
# AT4G18960.1




## Gene : **AG** *A. thaliana*

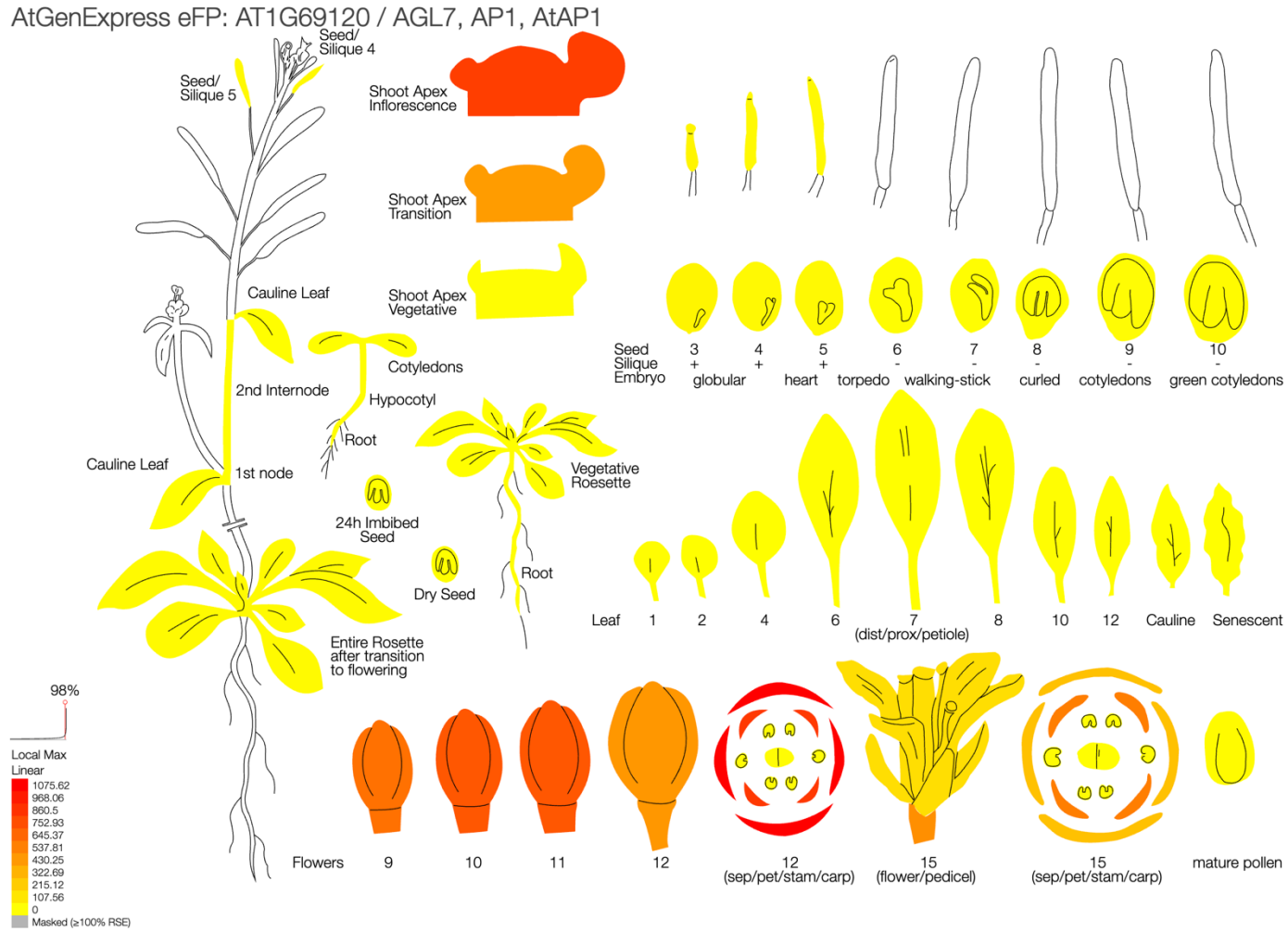
| | | | |
|---|---|---|---|
| DB identifier | **AT4G18960** | Secondary Identifier | **locus:2117144** |
| Name | **AGAMOUS** | Brief Description | **K-box region and MADS-box transcription factor family protein** |

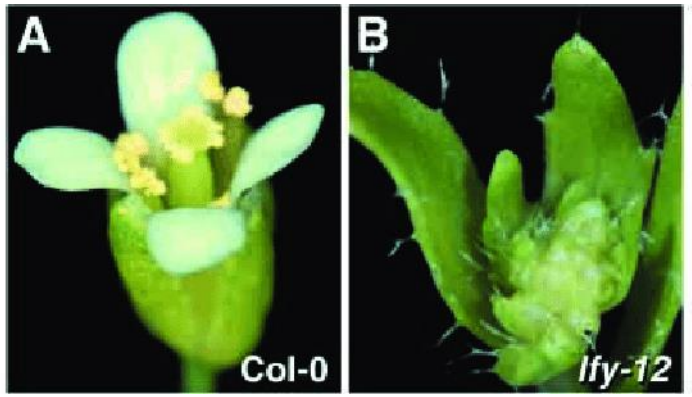
TAIR Computational Description **K-box region and MADS-box transcription factor family protein;(source:Araport11)**

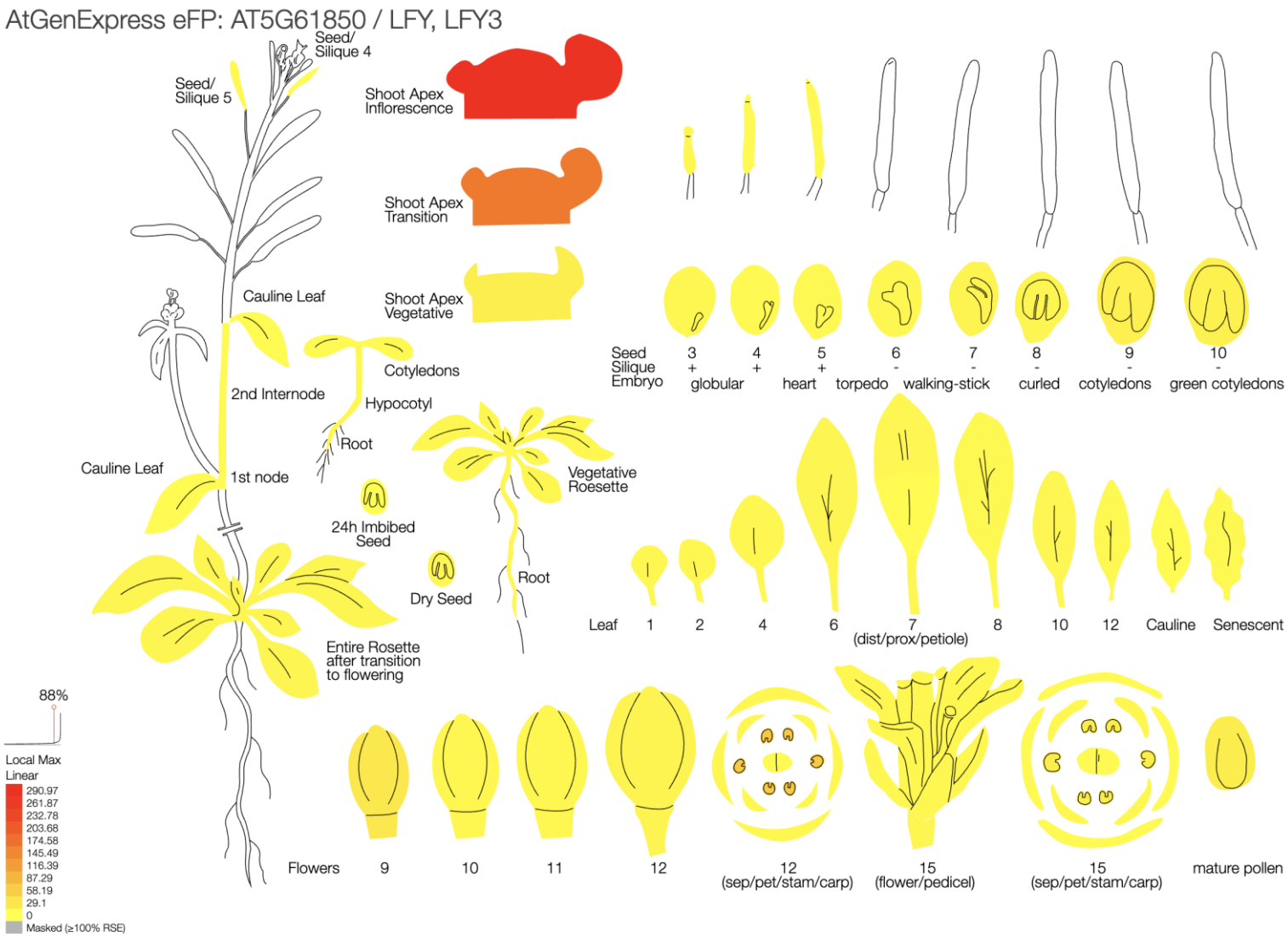



D
WT


A
ap1-1


AtGenExpress eFP: AT1G69120 / AGL7, AP1, AtAP1
Seed/ Silique 4
Seed/ Silique 5
Shoot Apex Inflorescence
Shoot Apex Transition
Shoot Apex Vegetative
Cauline Leaf
Cotyledons
2nd Internode
Hypocotyl
Root
Cauline Leaf
1st node
24h Imbibed Seed
Dry Seed
Vegetative Rosette
Root
Entire Rosette after transition to flowering
Seed Silique Embryo
3 + globular
4 + 
5 + heart
6 - torpedo
7 - walking-stick
8 - curled
9 - cotyledons
10 - green cotyledons
Leaf 1 2 4 6 7 (dist/prox/petiole) 8 10 12 Cauline Senescent
98%
Local Max
Linear
1075.62
968.06
860.5
752.93
645.37
537.81
430.25
322.69
215.12
107.56
0
Masked (≥100% RSE)
Flowers 9 10 11 12 12 (sep/pet/stam/carp) 15 (flower/pedicel) 15 (sep/pet/stam/carp) mature pollen


A
Col-0
B
lfy-12


AtGenExpress eFP: AT5G61850 / LFY, LFY3
Seed/
Silique 4
Seed/
Silique 5
Shoot Apex
Inflorescence
Shoot Apex
Transition
Shoot Apex
Vegetative
Cauline Leaf
Cotyledons
2nd Internode
Hypocotyl
Root
Cauline Leaf
1st node
24h Imbibed
Seed
Dry Seed
Vegetative
Roesette
Root
Entire Rosette
after transition
to flowering
Seed
Silique
Embryo
3
+
globular
4
+
5
+
heart
6
-
torpedo
7
-
walking-stick
8
-
curled
9
-
cotyledons
10
-
green cotyledons
Leaf
1
2
4
6
7
(dist/prox/petiole)
8
10
12
Cauline
Senescent
88%
Local Max
Linear
290.97
261.87
232.78
203.68
174.58
145.49
116.39
87.29
58.19
29.1
0
Masked (≥100% RSE)
Flowers
9
10
11
12
12
(sep/pet/stam/carp)
15
(flower/pedicel)
15
(sep/pet/stam/carp)
mature pollen

JOURNAL ARTICLE FEATURED

# How flower development genes were identified using forward genetic screens in *Arabidopsis thaliana*

David R Smyth ✉ Author Notes

*Genetics*, Volume 224, Issue 4, August 2023, iyad102,
https://doi.org/10.1093/genetics/iyad102
**Published:** 09 June 2023 **Article history** ▾

| Gene | Mutant Allele No.[a] | Encoded Protein[b] | References |
|---|---|---|---|
| *AGAMOUS (AG)* | 3, [4] | MADS TF (C) | Bowman *et al.* (1991), Sieburth *et al.* (1995) |
| *APETALA1 (AP1)* | [2–5, 7] | MADS TF (A) | Bowman *et al.* (1993) |
| *APETALA2 (AP2)* | 2, 8, 9 | AP2 TF (A) | Bowman *et al.* (1991) |
| *APETALA3 (AP3)* | None obtained | MADS TF (B) | Bowman *et al.* (1991) |
| *CAULIFLOWER (CAL)* | [1[c]] | MADS TF | Bowman *et al.* (1993) |
| *CLAVATA1 (CLV1)* | S292, S368 | LRR-RLK | unpublished |
| *CLAVATA3 (CLV3)* | 1, [2[d]] | CLE signaling peptide | Clark *et al.* (1995) |
| *CRABS CLAW (CRC)* | 1, [2[d]] | YABBY TF | Alvarez and Smyth (1999, 2002), Bowman and Smyth (1999) |
| *LEAFY (LFY)* | 3, 4 [5, 6, 10[e], 11] | LEAFY TF | Weigel *et al.* (1992) |
| *LEUNIG (LUG)* | 8, 9 | TUP1 co-repressor | Liu and Meyerowitz (1995) |
| *PETAL LOSS (PTL)* | [1[e], 3–5] | Trihelix TF | Griffith *et al.* (1999), Brewer *et al.* (2004) |
| *PETAL LOSS MODIFIER (PMD)* | [1[f]] | unknown | Griffith *et al.* (1999) |
| *PIN FORMED1 (PIN1)* | [3, 5] | Auxin efflux carrier | Bennett *et al.* (1995) |
| *PINOID (PID)* | 1, [2, 3[e], 4, 7] | S/T protein kinase | Bennett *et al.* (1995) |
| *PISTILLATA (PI)* | 2, 3 | MADS TF (B) | Bowman *et al.* (1991) |
| *SEPALLATA1-4 (SEP1-4)* | None obtained | MADS TF (E) | |
| *SPATULA (SPT)* | 1, [2, 3] | bHLH TF | Alvarez and Smyth (1999, 2002), Heisler *et al.* (2001) |
| *SUPERMAN (SUP)* | [4[e]] | C2H2 ZnF TF | Bowman *et al.* (1992) |
| *TERMINAL FLOWER1 (TFL1)* | 2–5, [6, 7, 8[e]] | CETS | Alvarez *et al.* (1992) |
| *UNUSUAL FLORAL ORGANS (UFO)* | [3–5] | F-box | Levin and Meyerowitz (1995) |



# Mutate the mutant!

EMS

M1 -------> M2





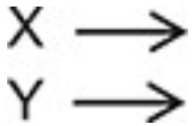
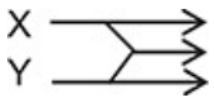
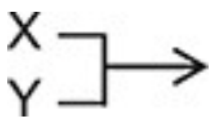


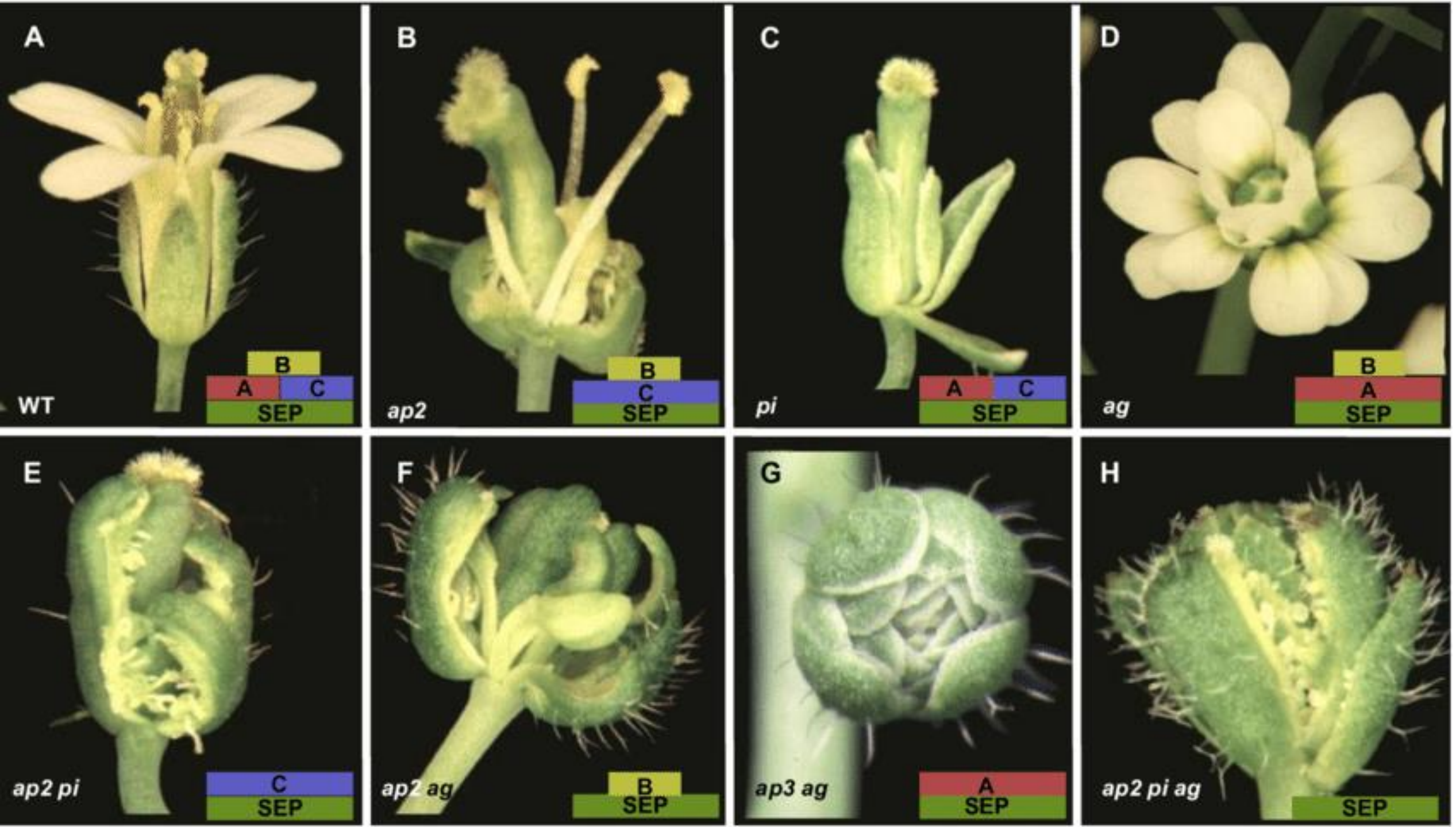


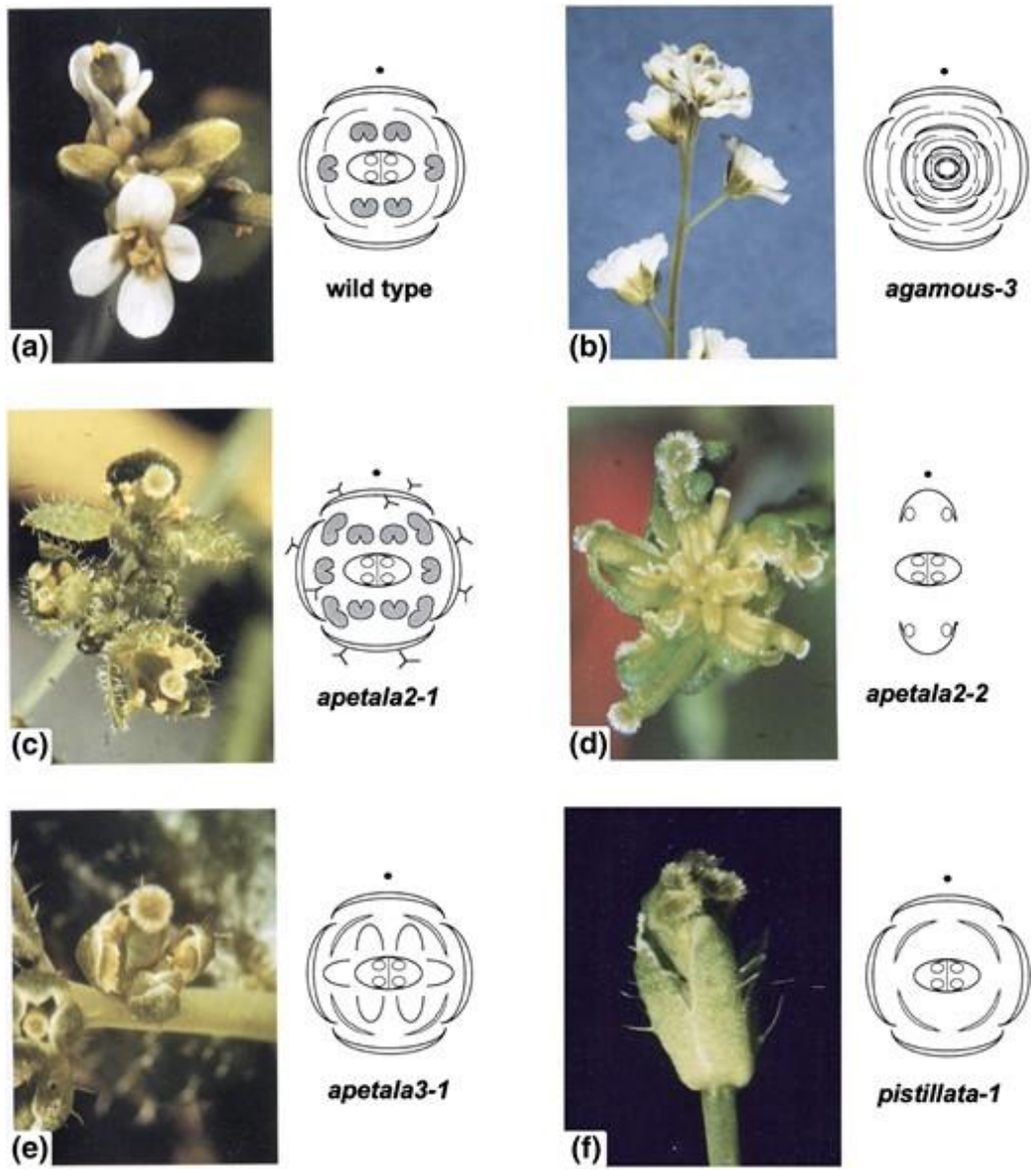


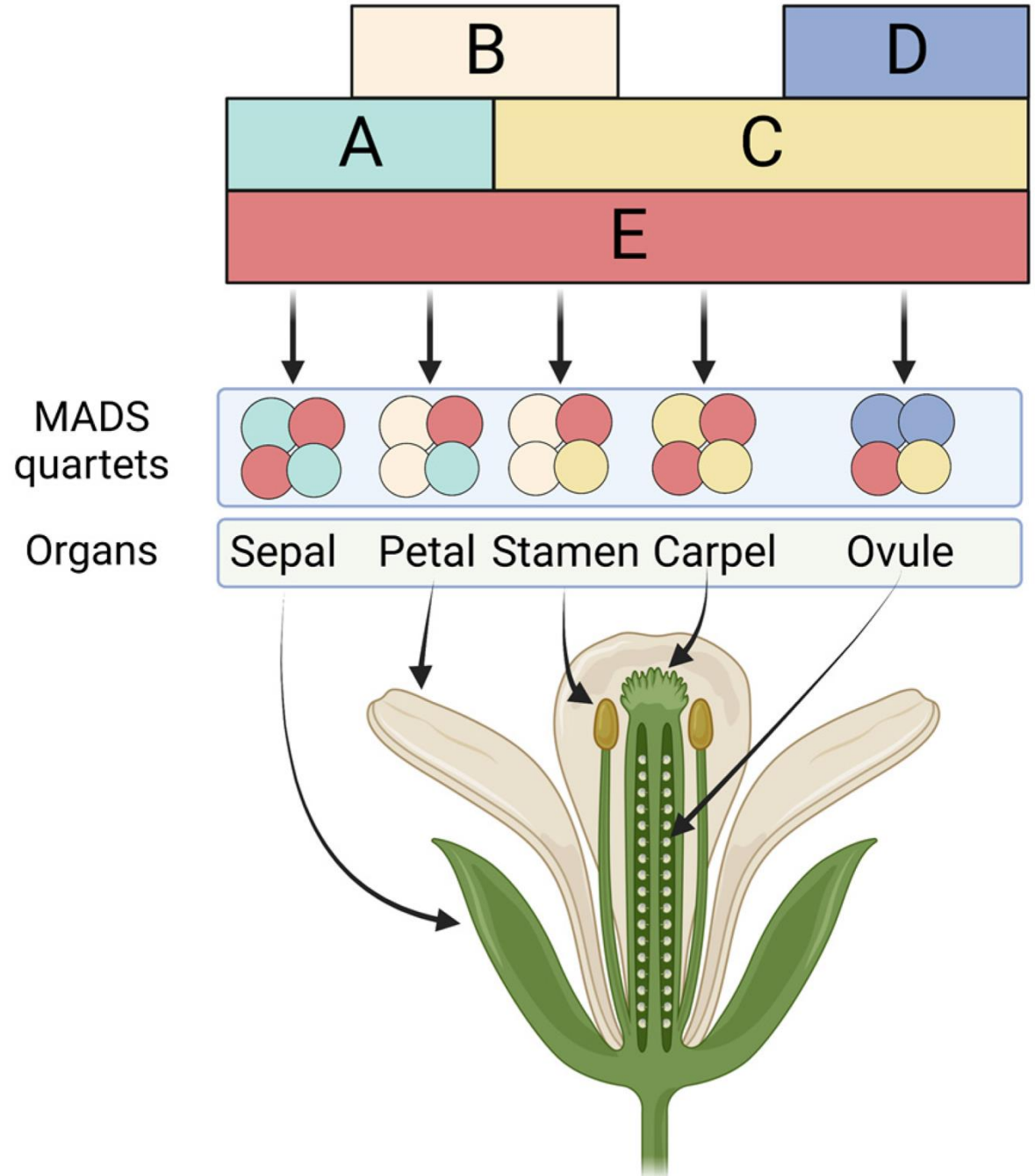
**Reversion!**

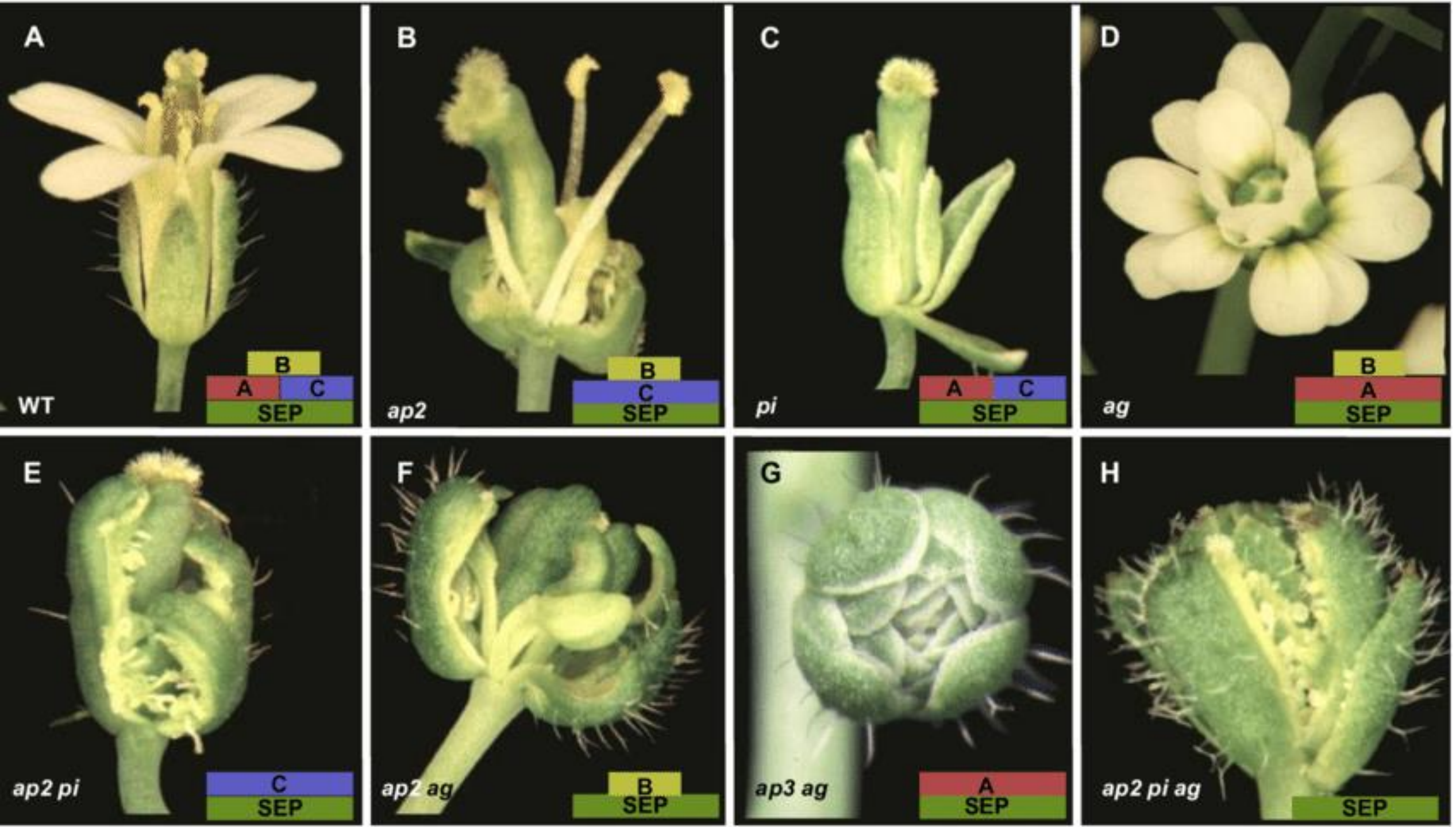


| Class of interaction | | Double mutant phenotype | Deduction |
|---|---|---|---|
| Additive |  | Double mutant combines phenotypes of single mutants | X and Y have different roles |
| Overlapping |  | Double mutant is more disrupted than the additive phenotype | X and Y share part function |
| Epistatic |  | Double and X mutant the same, Y mutant different | X acts upstream of Y |
| Shared |  | Double mutant the same as both X and Y single mutants | Both X and Y are required |
| Redundant |  | Double mutant is disrupted, X and Y single mutants wild type | Either X or Y is required |


A
WT
B
A
C
SEP
B
ap2
B
C
SEP
C
pi
A
C
SEP
D
ag
B
A
SEP
E
ap2 pi
C
SEP
F
ap2 ag
B
SEP
G
ap3 ag
A
SEP
H
ap2 pi ag
SEP

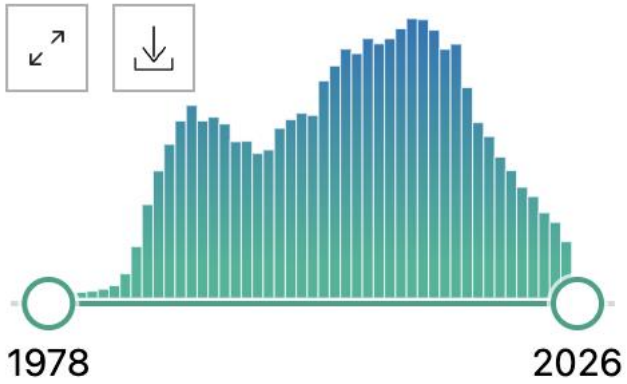

(a)
wild type
(b)
agamous-3
(c)
apetala2-1
(d)
apetala2-2
(e)
apetala3-1
(f)
pistillata-1

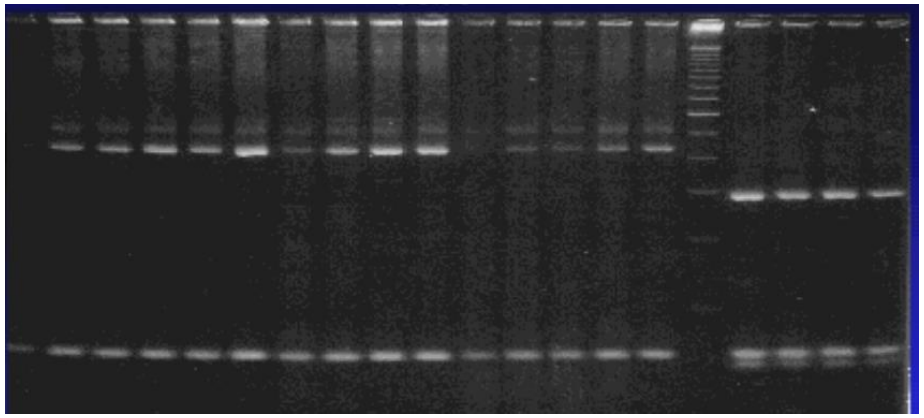

B
D
A
C
E
MADS quartets
Organs
Sepal
Petal
Stamen
Carpel
Ovule


A
WT
B
A
C
SEP
B
ap2
B
C
SEP
C
pi
A
C
SEP
D
ag
B
A
SEP
E
ap2 pi
C
SEP
F
ap2 ag
B
SEP
G
ap3 ag
A
SEP
H
ap2 pi ag
SEP

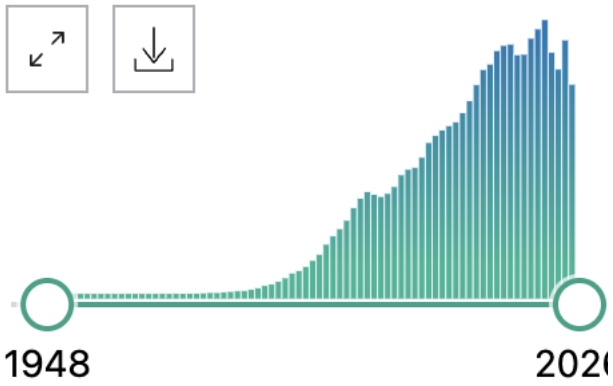

KpnI
Sau3AI
MseI
RsaI
SspI
TaqI
MW
MW
AluI
BamHI
BfaI
BstUI
DraI
EcoRI
CauPh-YNym
B

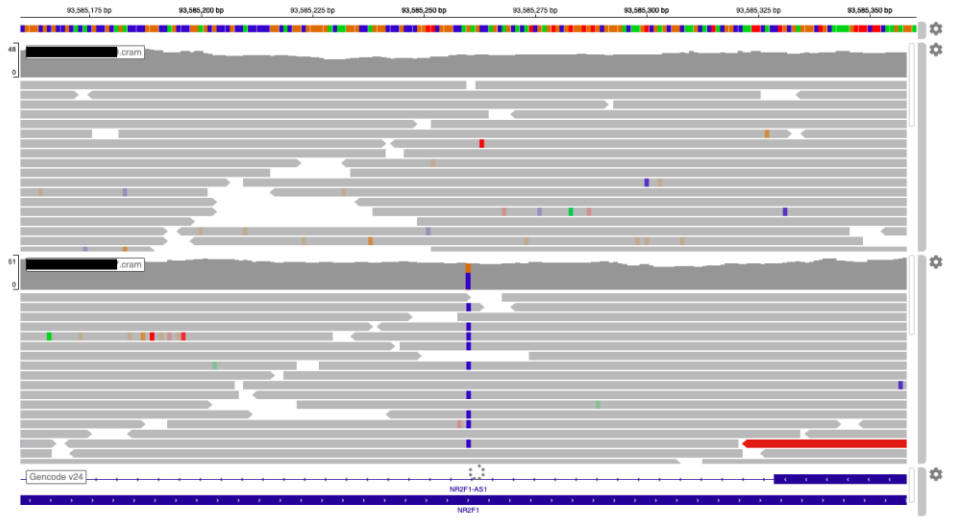
*RFLP Polymorphism*



*Genome sequencing*



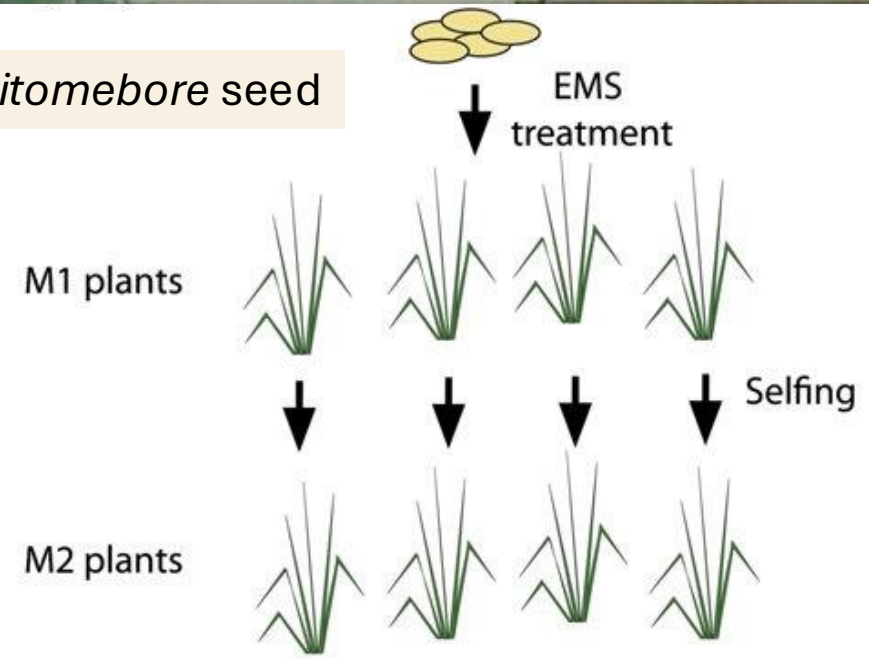




味蔵
宮古店
テレトラック
MIYAKO





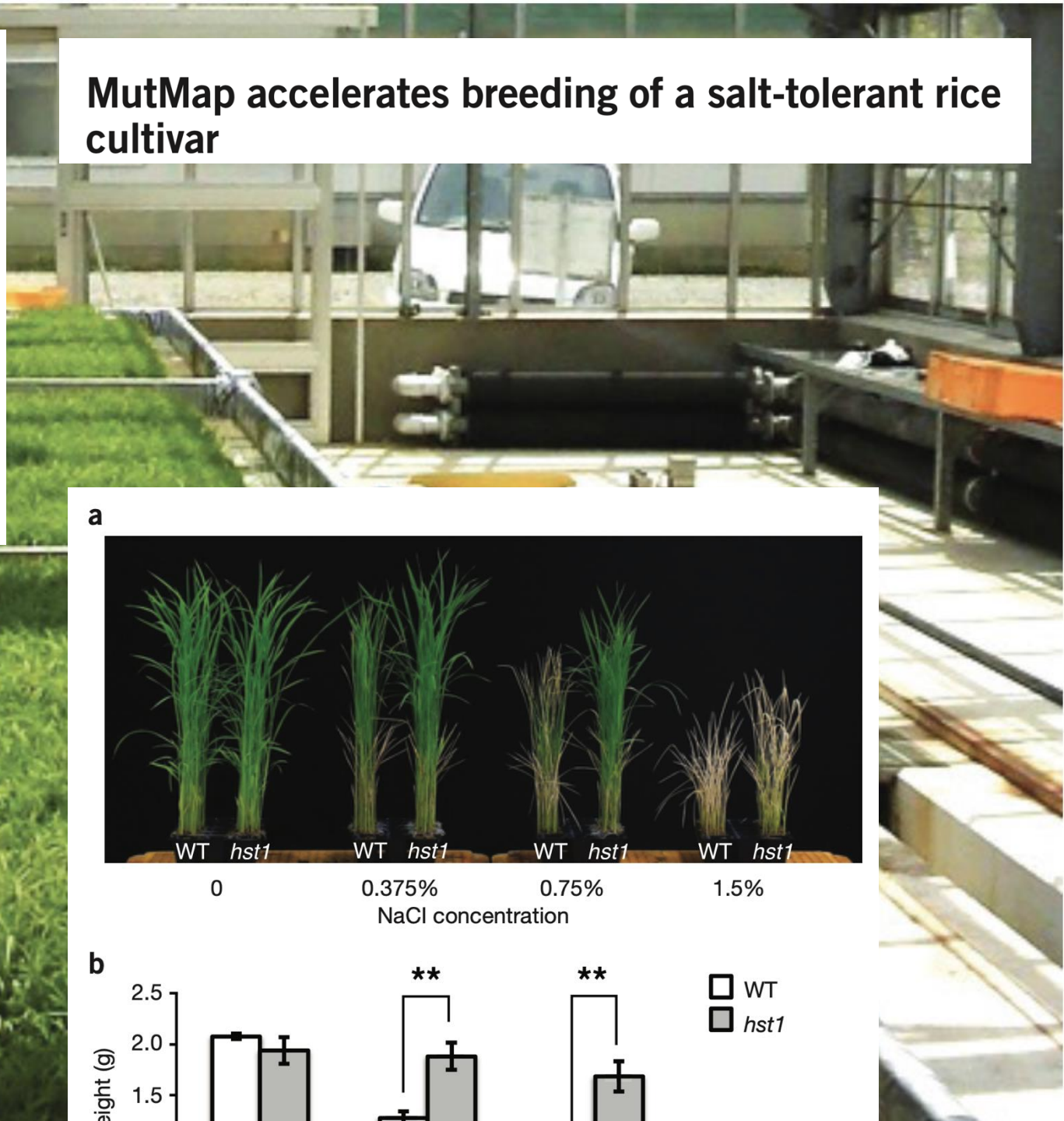
*Hitomebore* seed



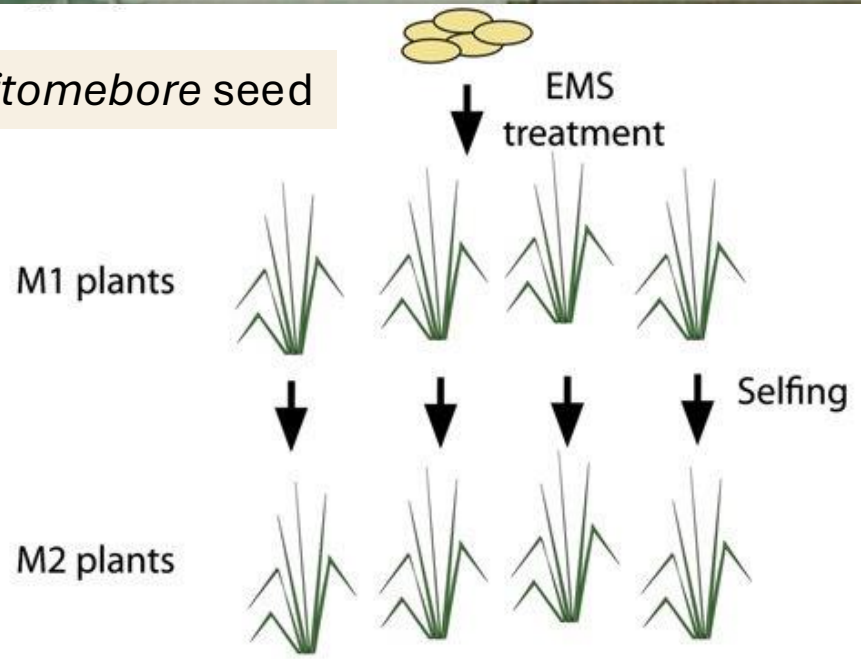

2015

# MutMap accelerates breeding of a salt-tolerant rice cultivar



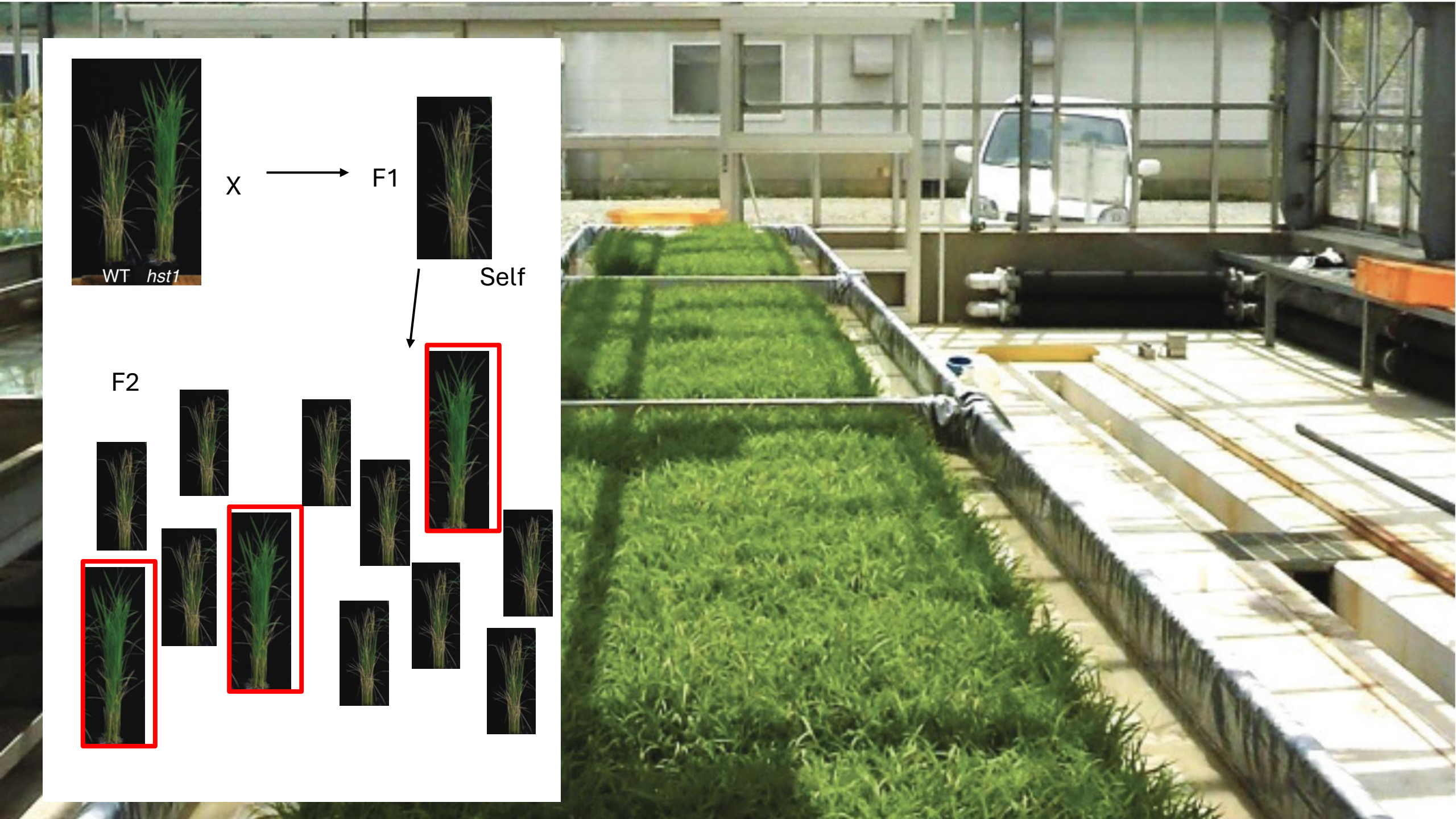


*Hitomebore* seed




2015

## MutMap accelerates breeding of a salt-tolerant rice cultivar


WT
hst1
X
F1
Self
F2

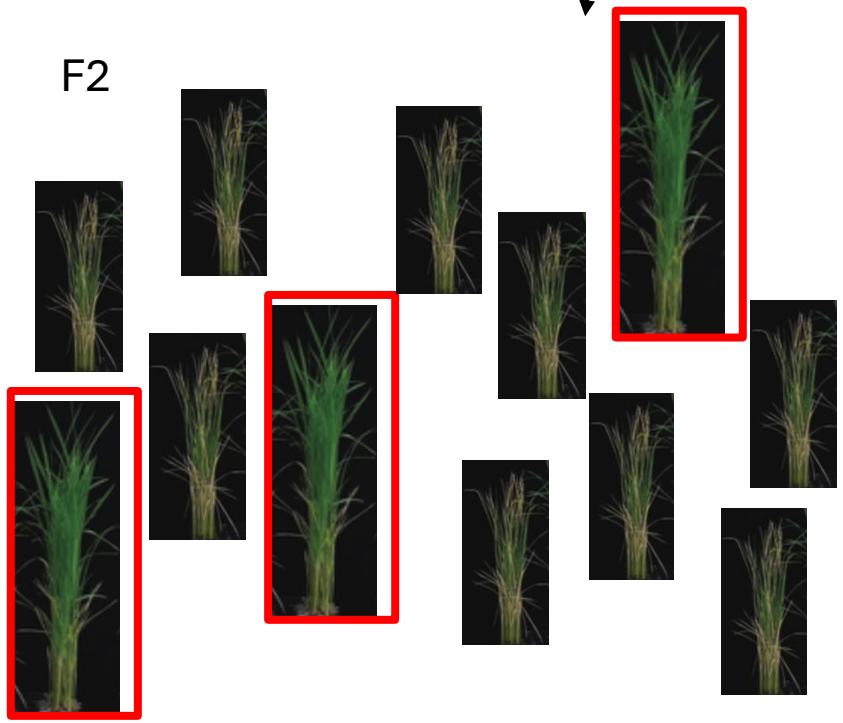



X

F1

Self

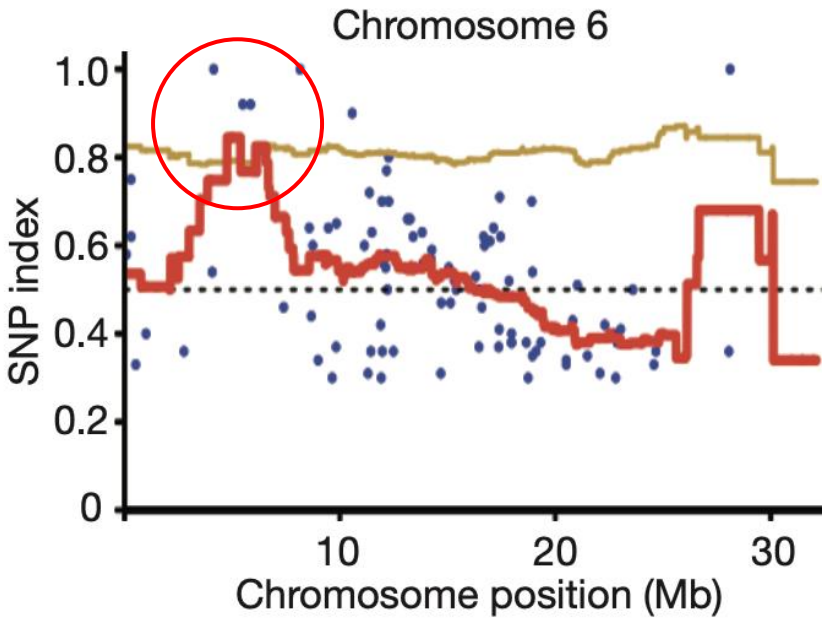
F2



F2



Pooled DNA sequencing





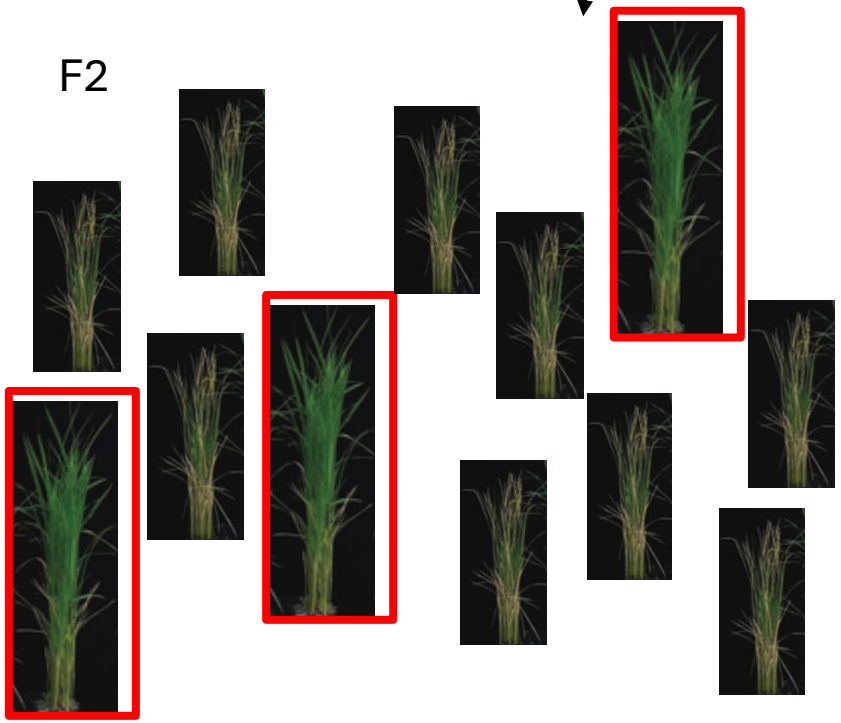

X

F1



Self

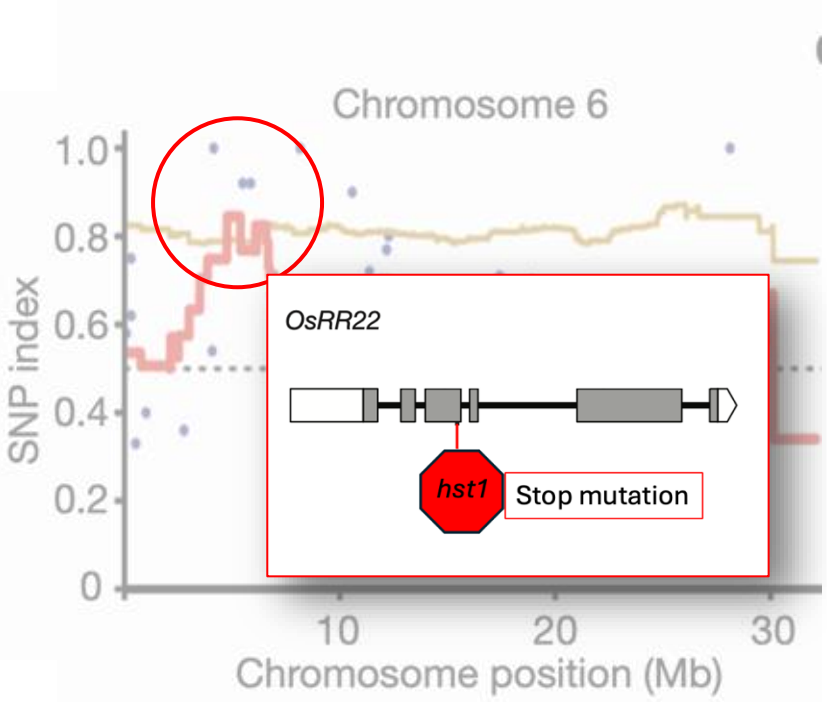
F2

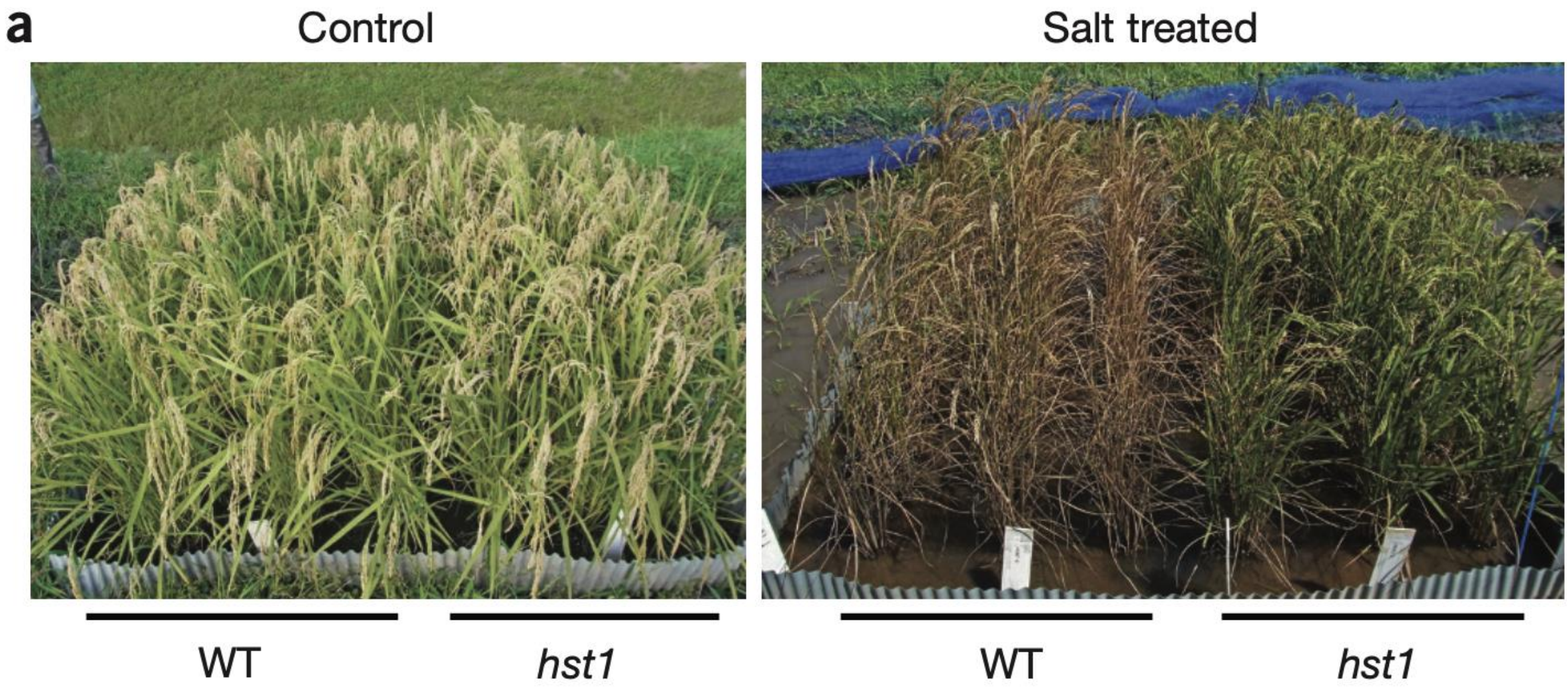


F2



Pooled DNA sequencing

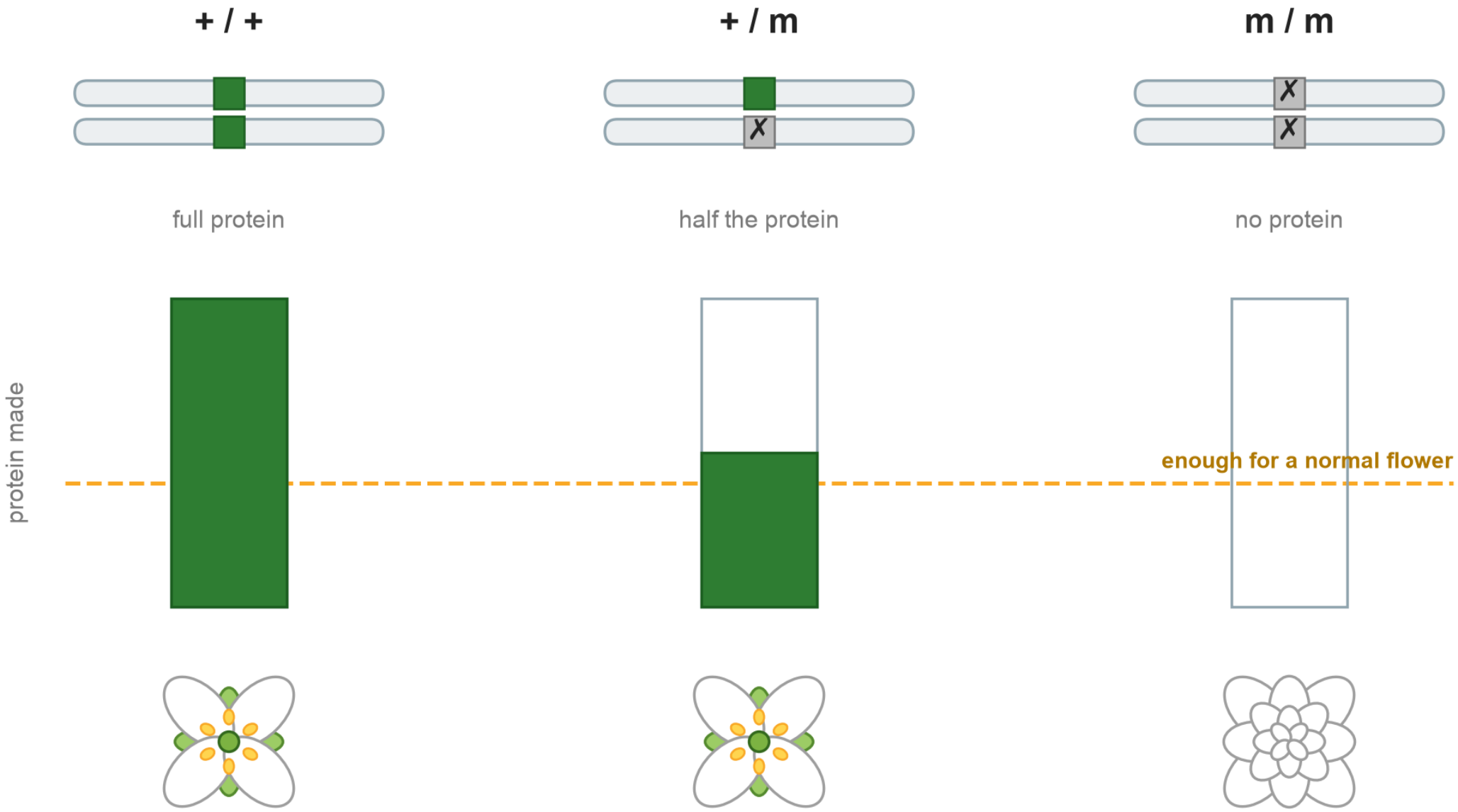



a
Control
Salt treated
WT
hst1
WT
hst1

***Factors to consider for mutant screen with mutagenesis***

How many mutations will arise in M1 from mutagenesis treatment?
How many mutations do you expect could produce your phenotype?
Do you expect new mutations to be dominant or recessive?

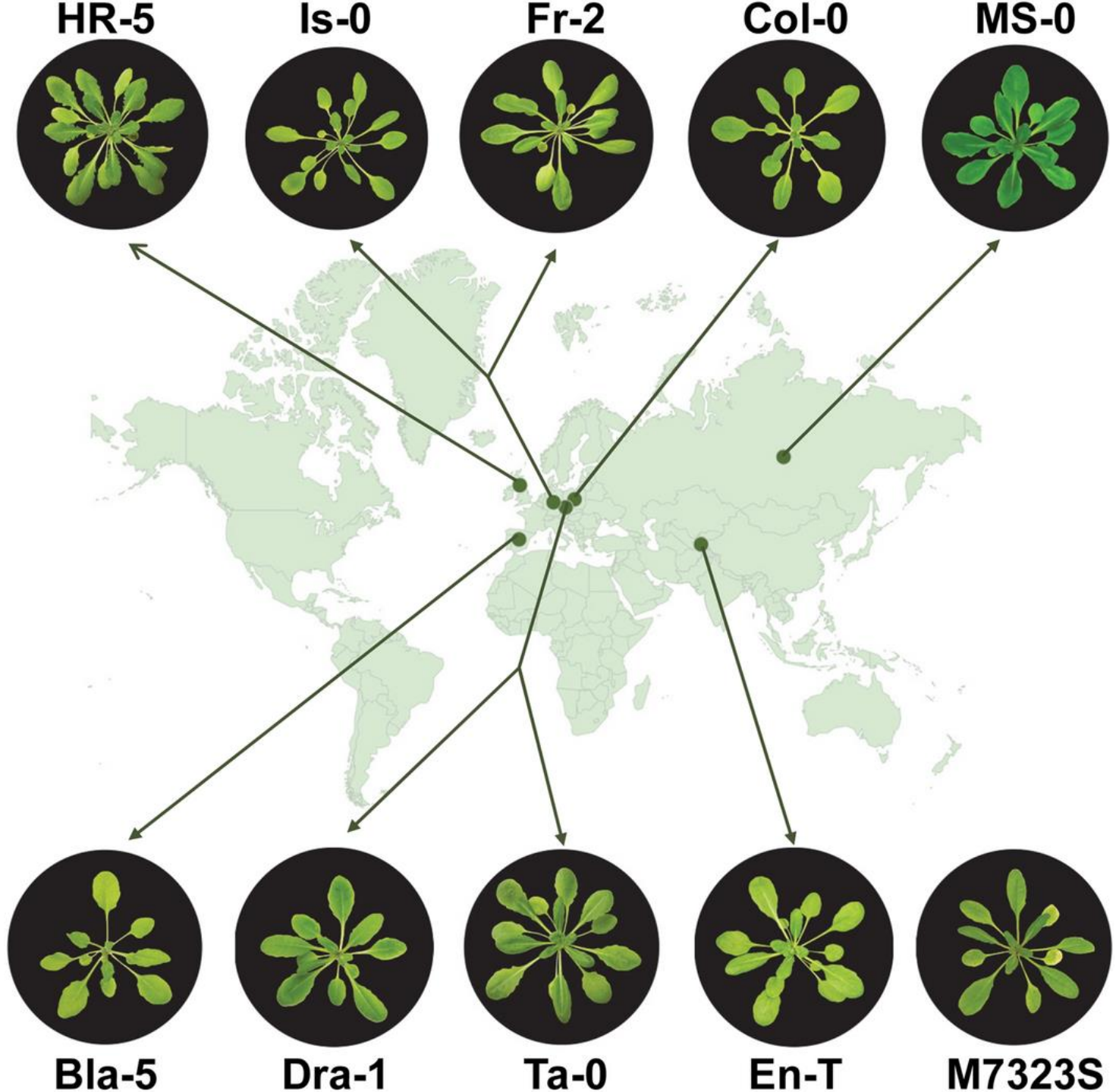
# New knockouts: dominant or recessive?

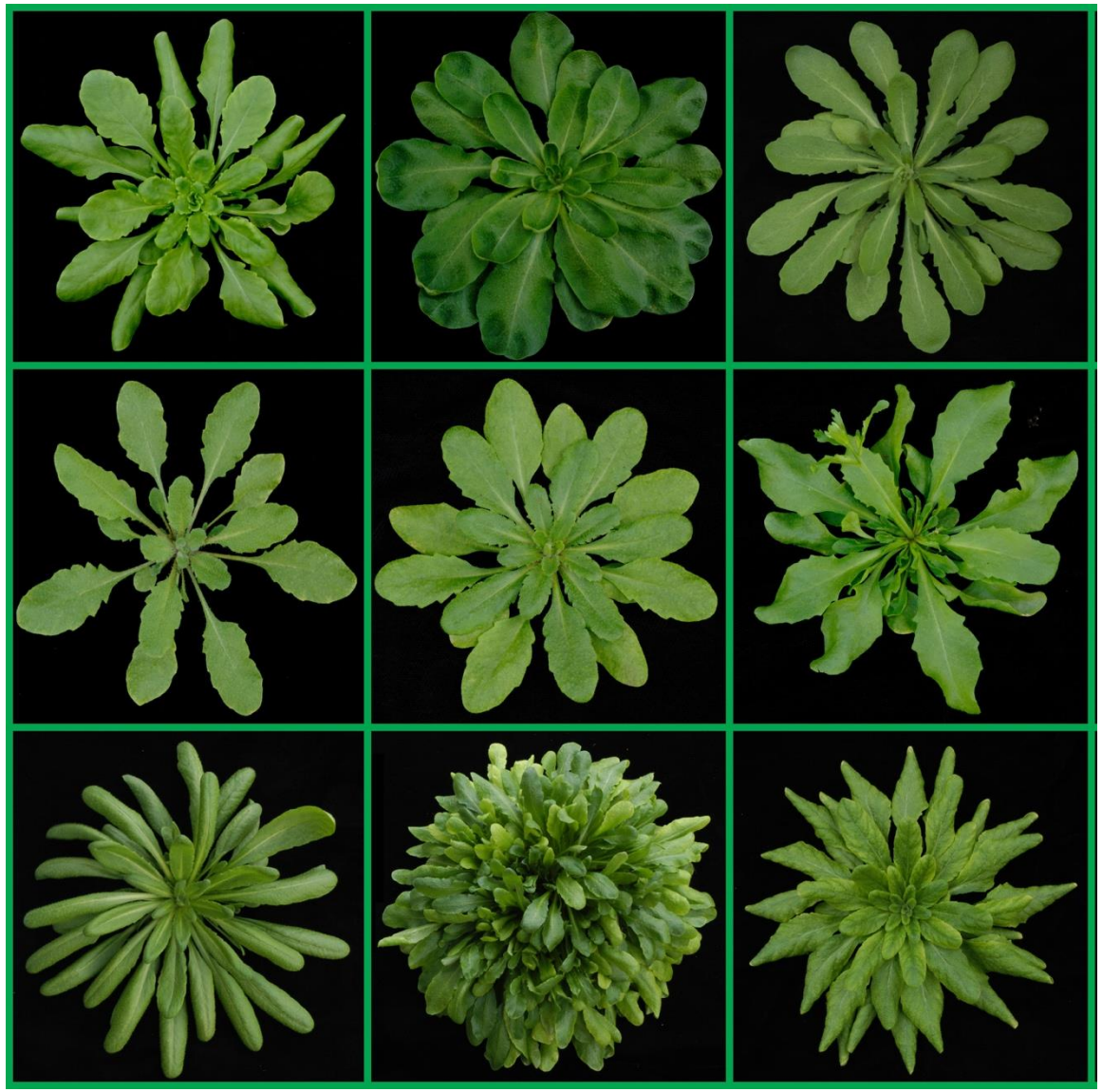


# Part 2: nature's mutants

Mapping natural variation: QTL, then GWAS

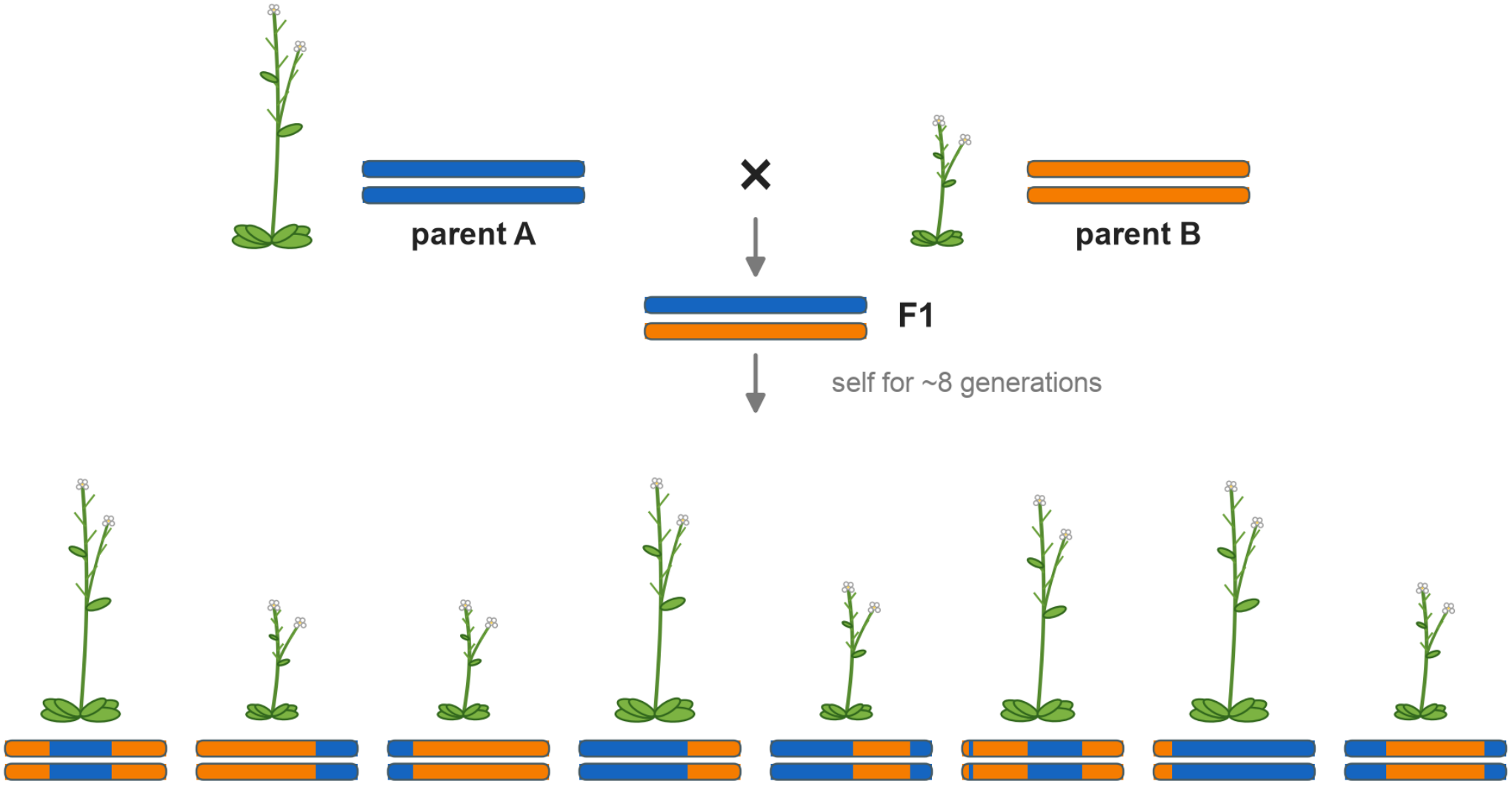



HR-5
Is-0
Fr-2
Col-0
MS-0
Bla-5
Dra-1
Ta-0
En-T
M7323S



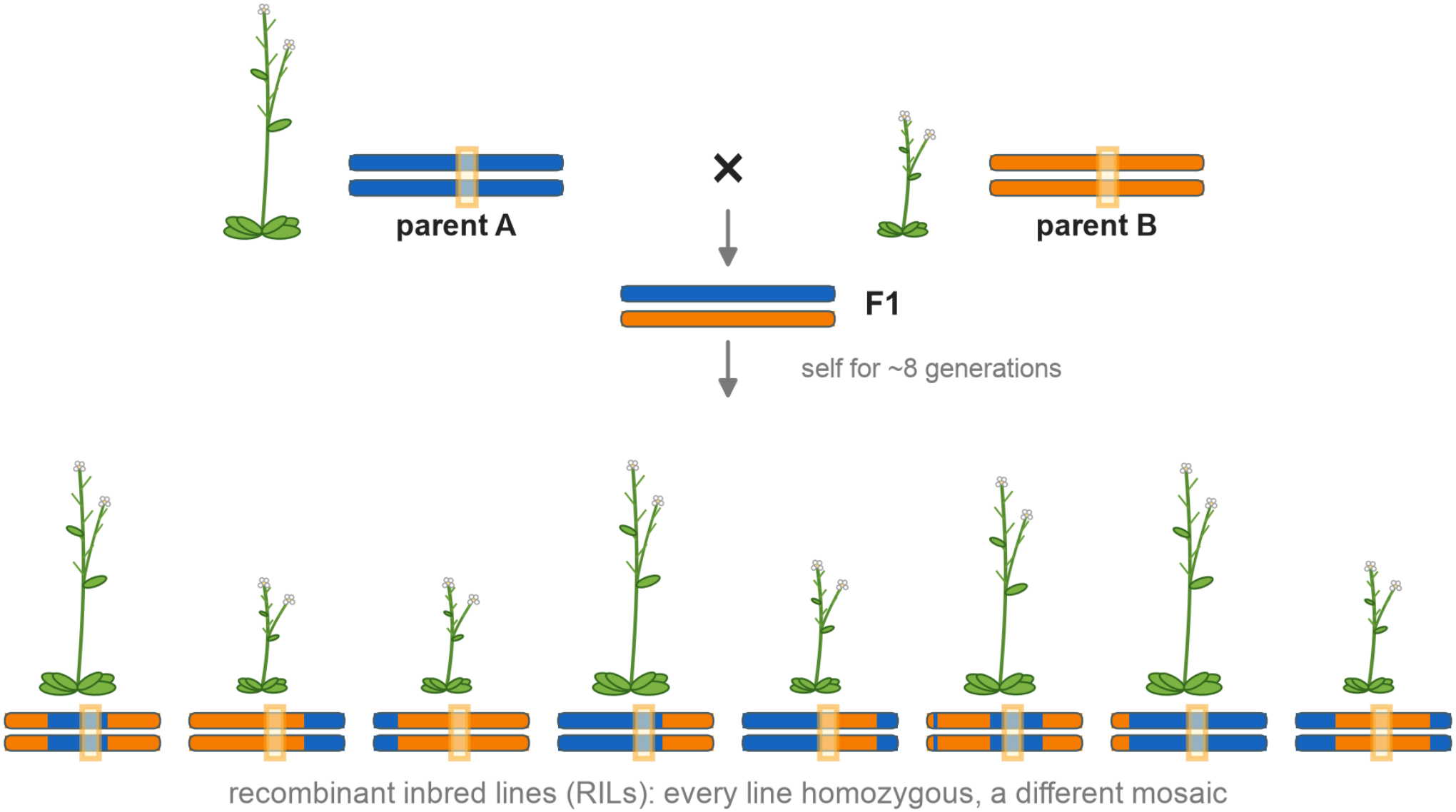


# QTL mapping: make your own population

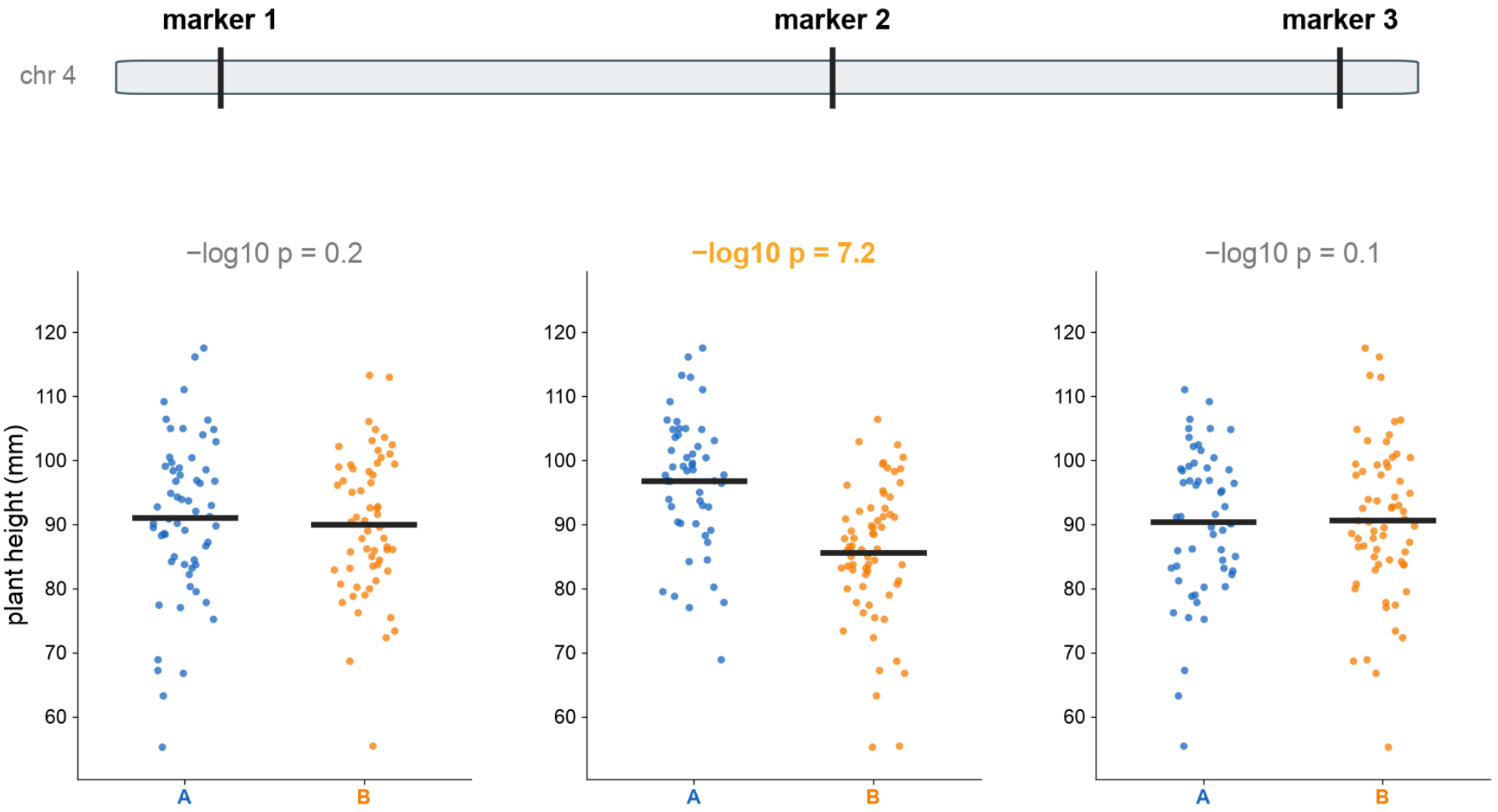



recombinant inbred lines (RILs): every line homozygous, a different mosaic

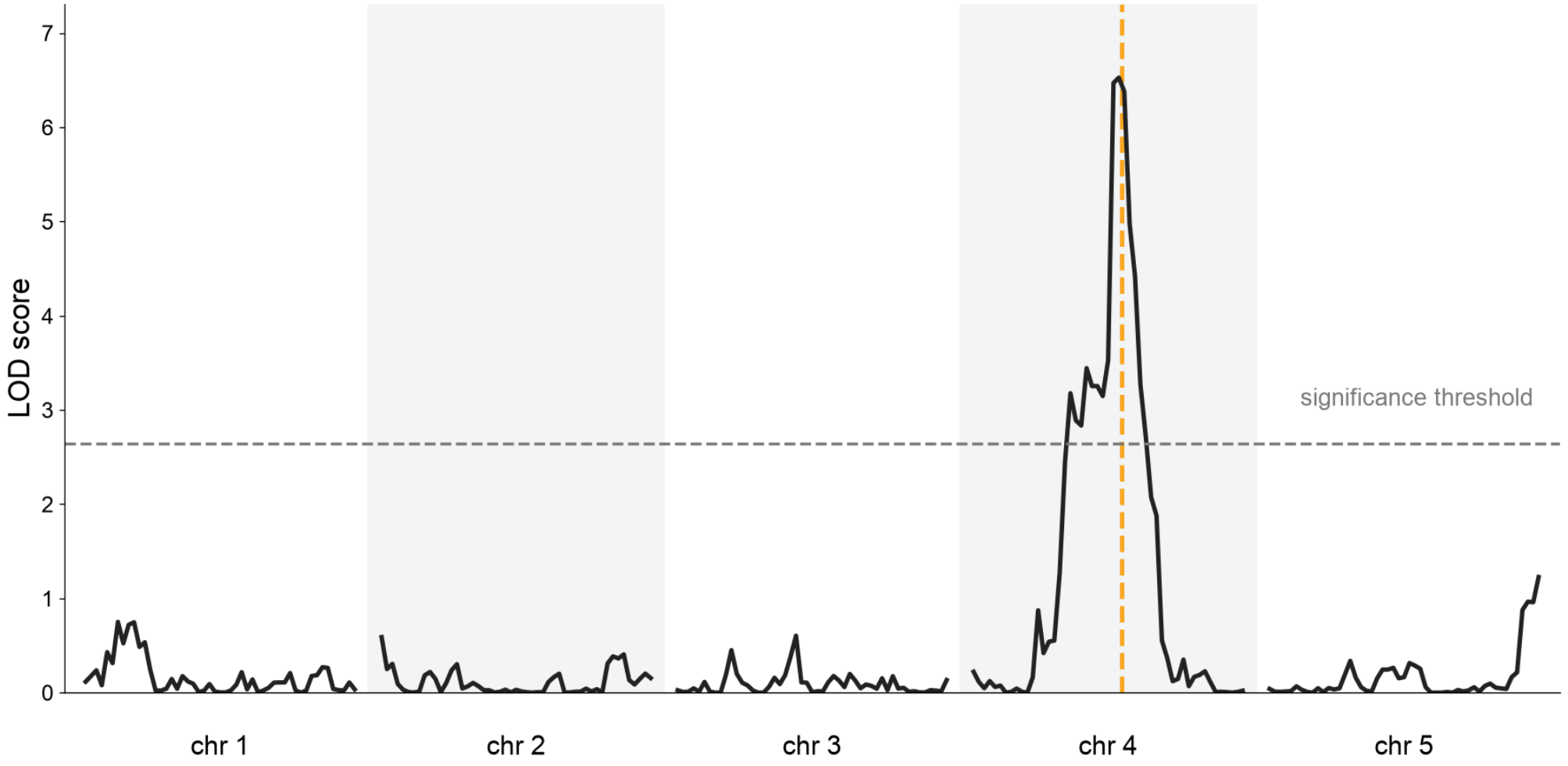
# What do the tall lines share?



# Test every marker



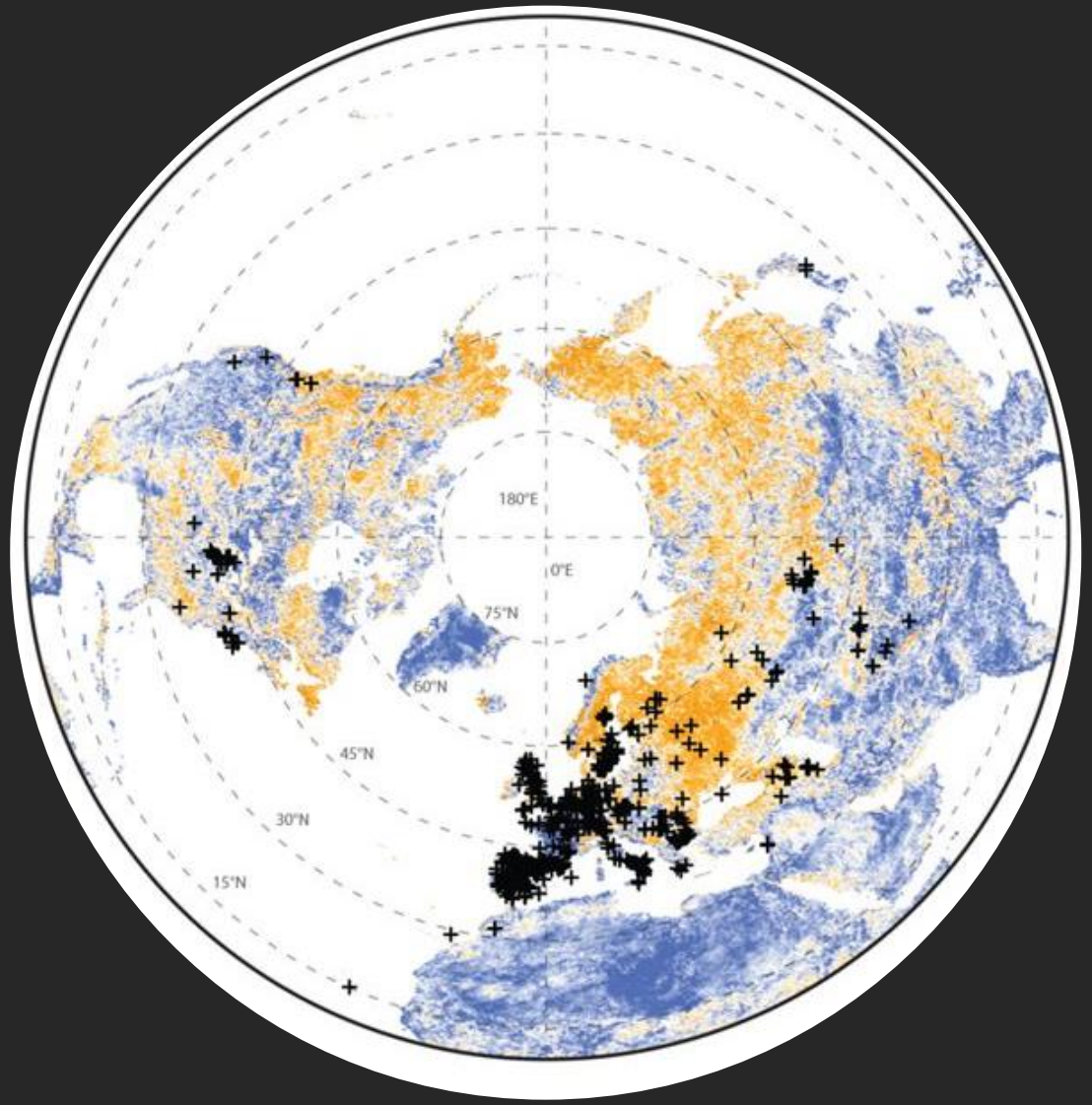
# Scan the genome → a QTL



# 2000: first *Arabidopsis thaliana* genome was sequenced



# Now: >1000 genomes sequenced








| LOCUS | Genome position | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | Chromosome 1: 113049 | C | A | A | A | C | C | C |
| 2 | Chromosome 1:115287 | T | T | T | G | G | G | G |
| 3 | Chromosome 1:117868 | G | T | G | T | G | G | T |
| | ... | | | | | | | |

**Millions of sites x**
**Hundreds-thousands of genotypes**

# Live poll

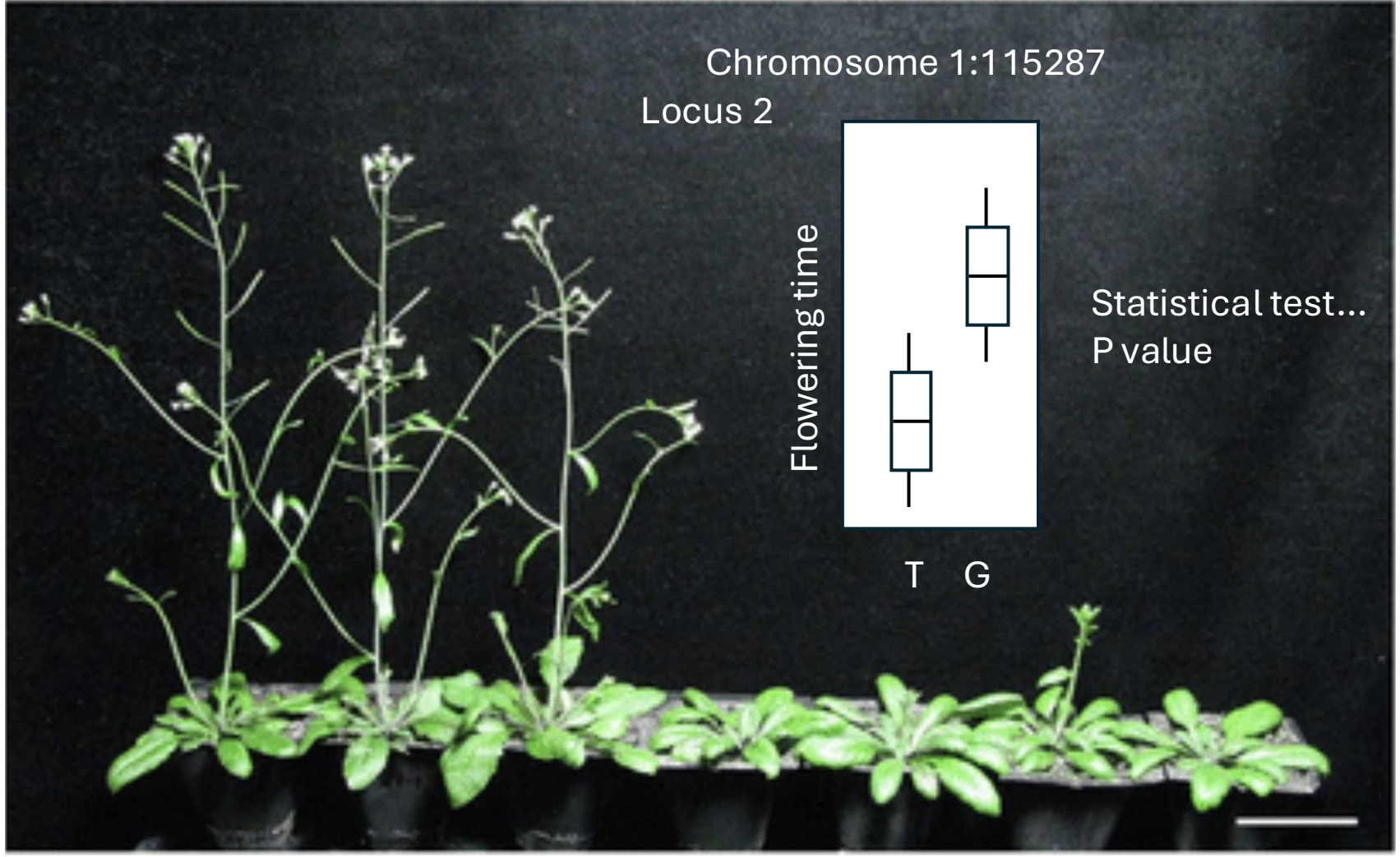
**Multiple choice**

## Which locus do you predict affects flowering time the most?

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-fg-locus

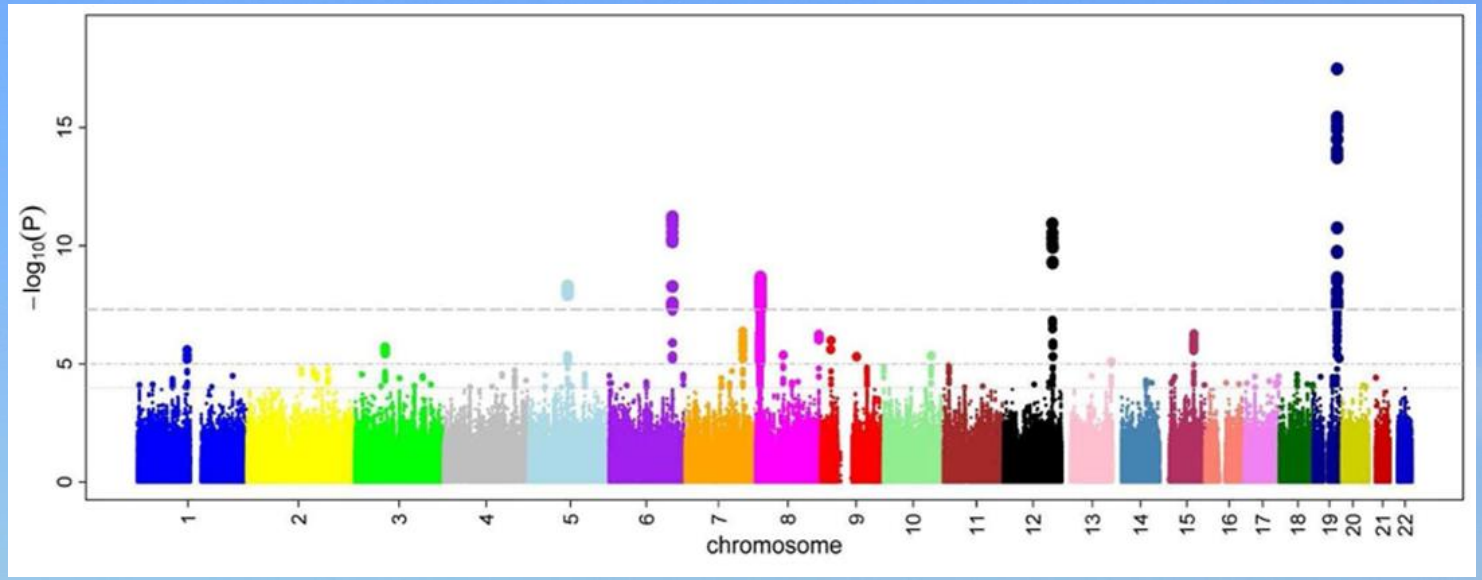
host screen




| LOCUS | Genome position | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | Chromosome 1: 113049 | C | A | A | A | C | C | C |
| 2 | Chromosome 1:115287 | T | T | T | G | G | G | G |
| 3 | Chromosome 1:117868 | G | T | G | T | G | G | T |
| | ... | | | | | | | |

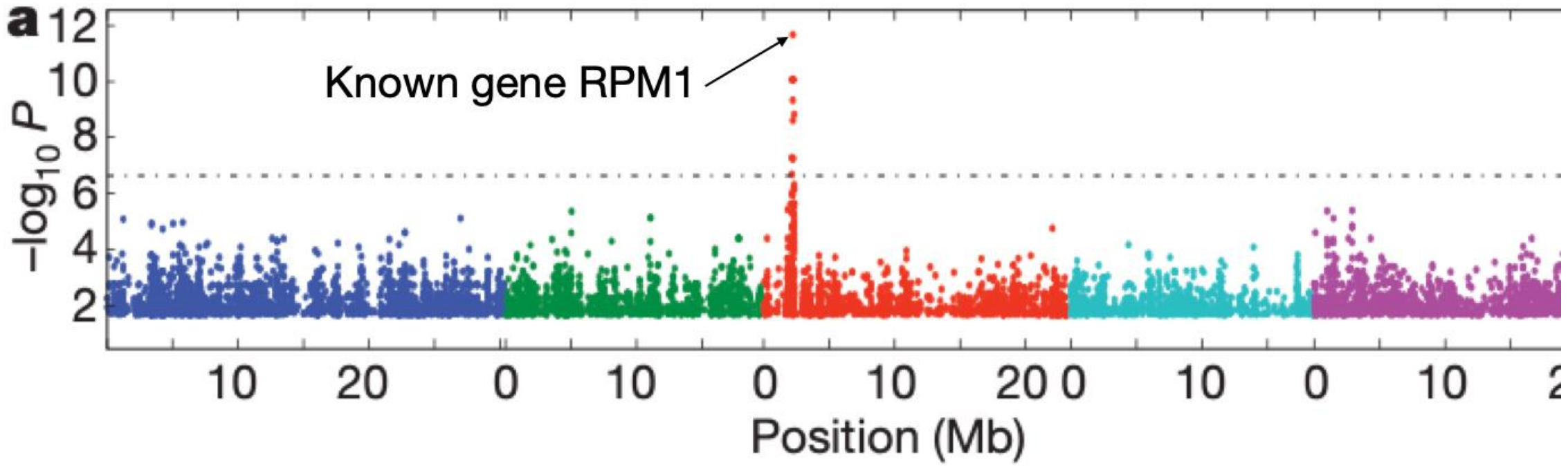
**Millions of sites x**
**Hundreds-thousands of genotypes**

Manhattan plot




Manhattan, New York

# Reading a Manhattan plot




**1 dot = 1 SNP**

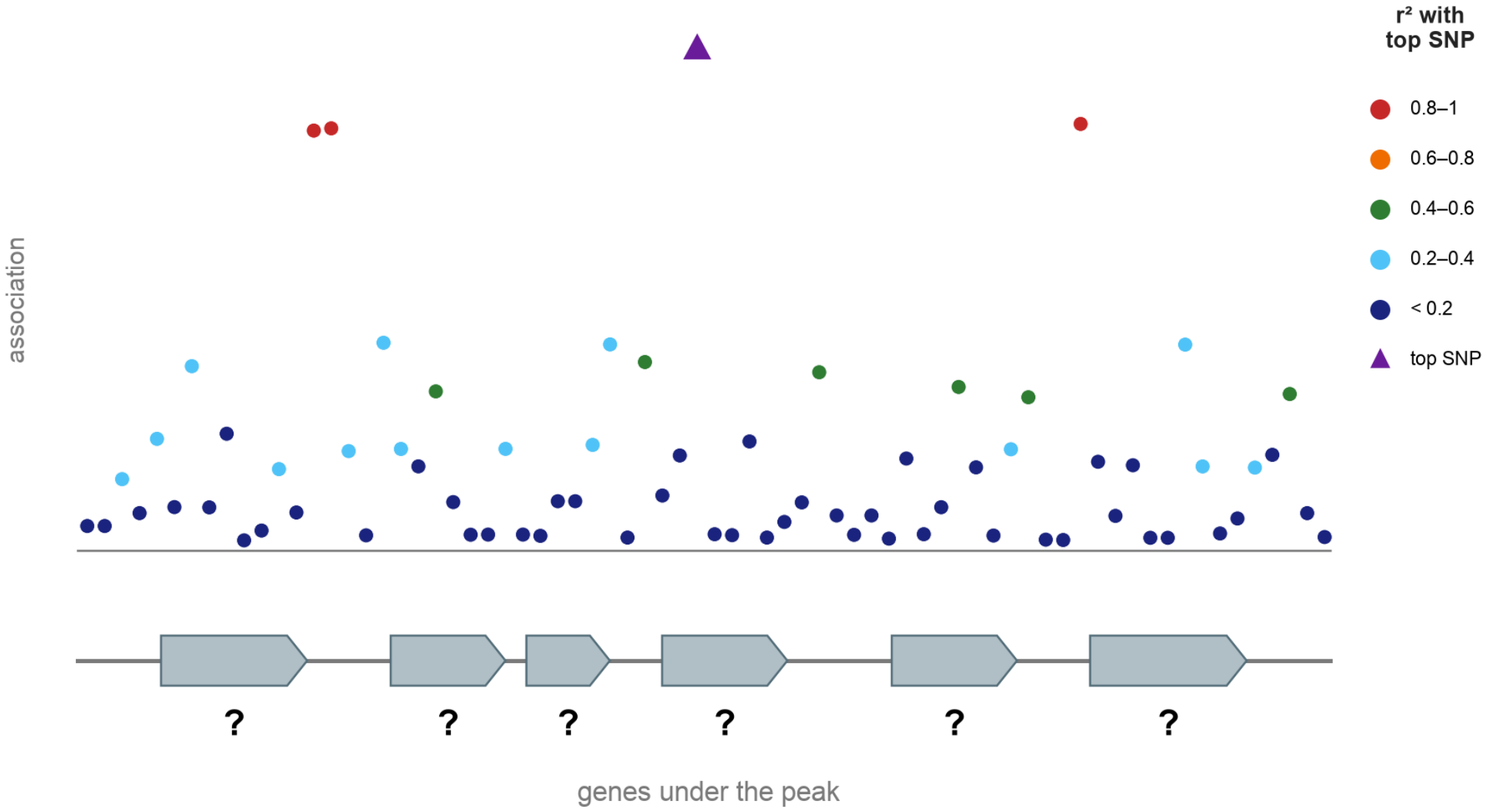
**higher = stronger association**

**dashed line = significance**

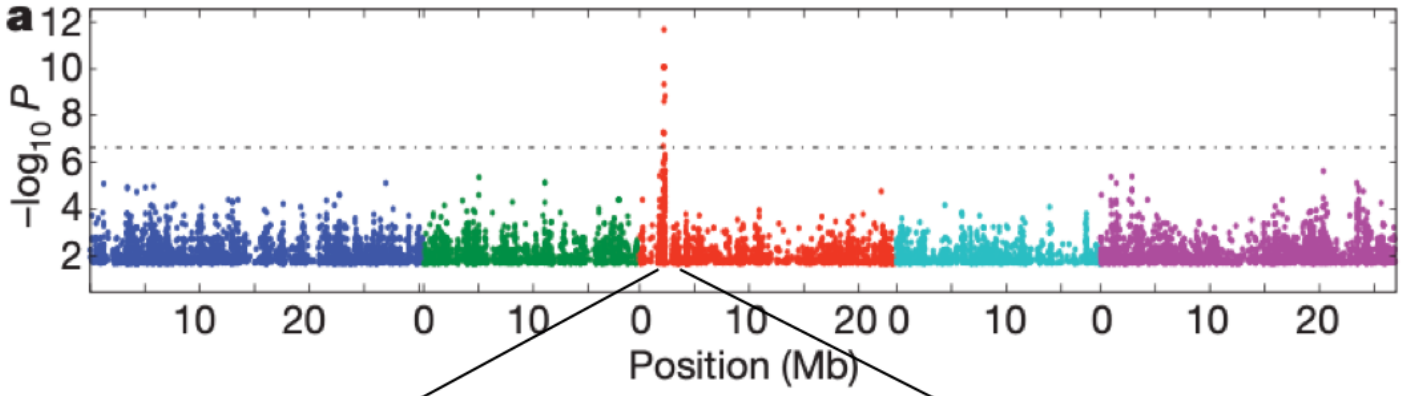
**5 colors = 5 chromosomes**

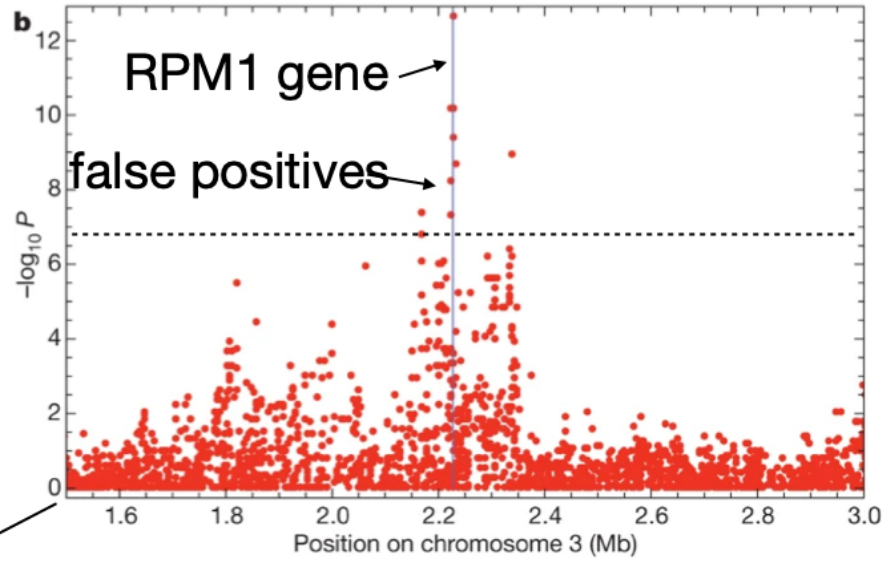
Atwell et al. 2010, Nature 465:627

# A peak is a region, not a gene



# Each “hit” has many associated markers



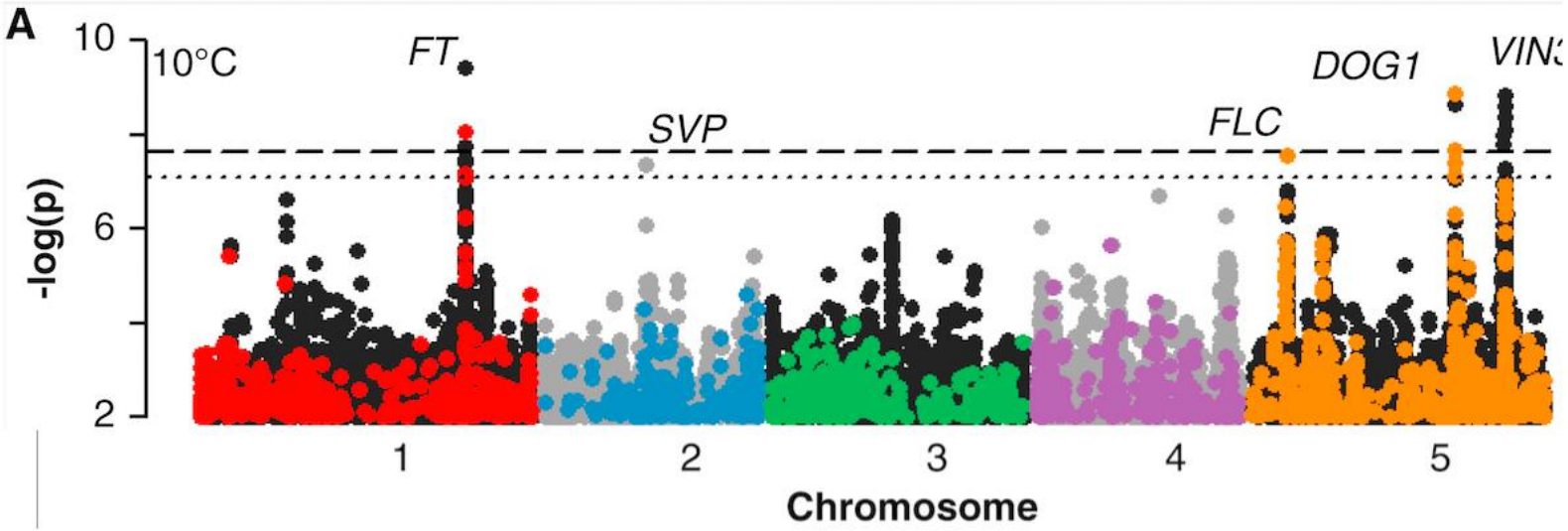







Gene positions

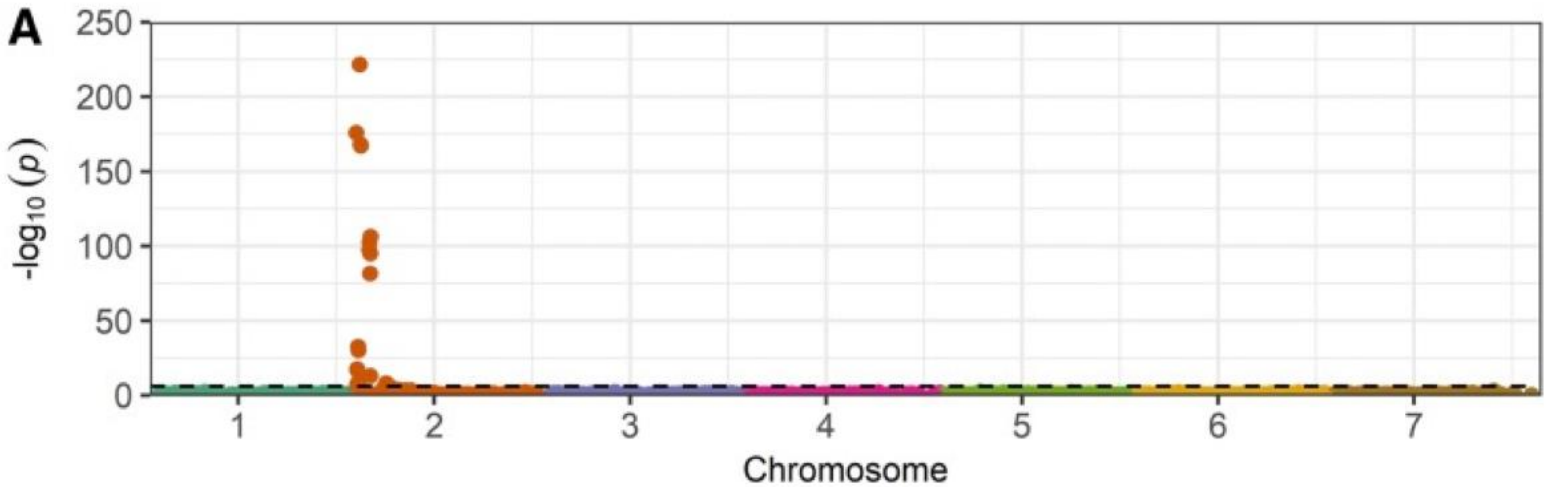
# Some examples of GWAS successes in plants




*Arabidopsis thaliana*

GWAS finds many known genes involved in flowering regulation

~1000 accessions




Strawberry

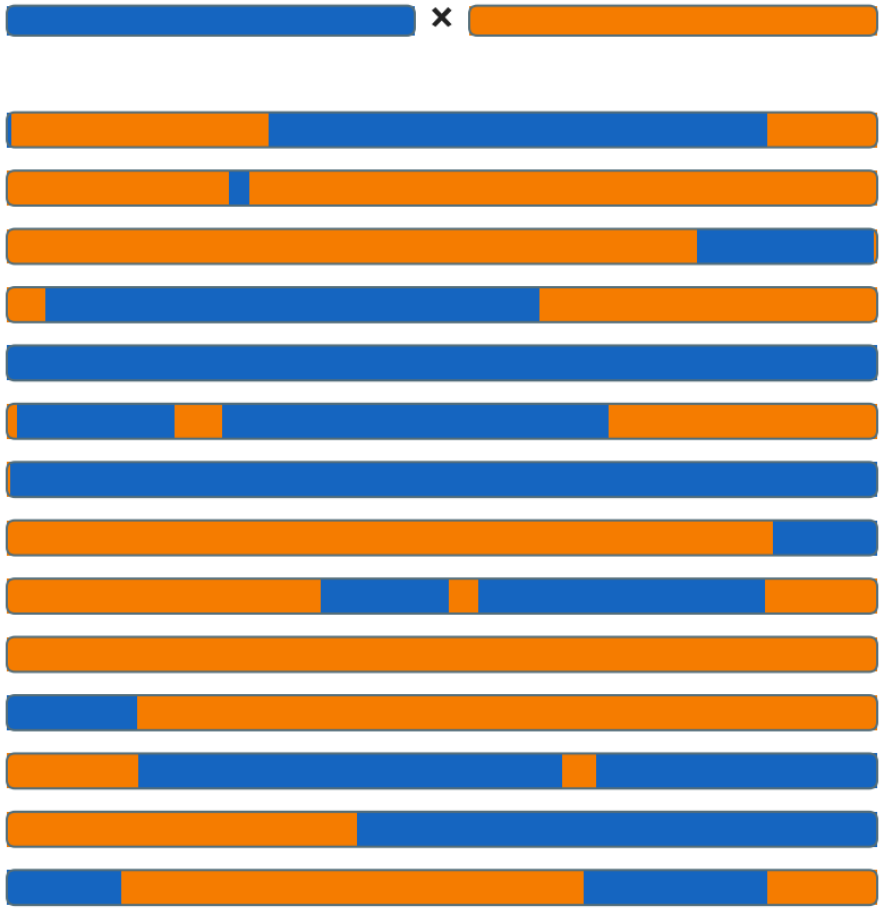
GWAS finds Fw1 locus for Fusarium Wilt Resistance

565 accessions

Currently used in breeding at UCD

# Why the two populations look so different

## Biparental cross (QTL mapping)



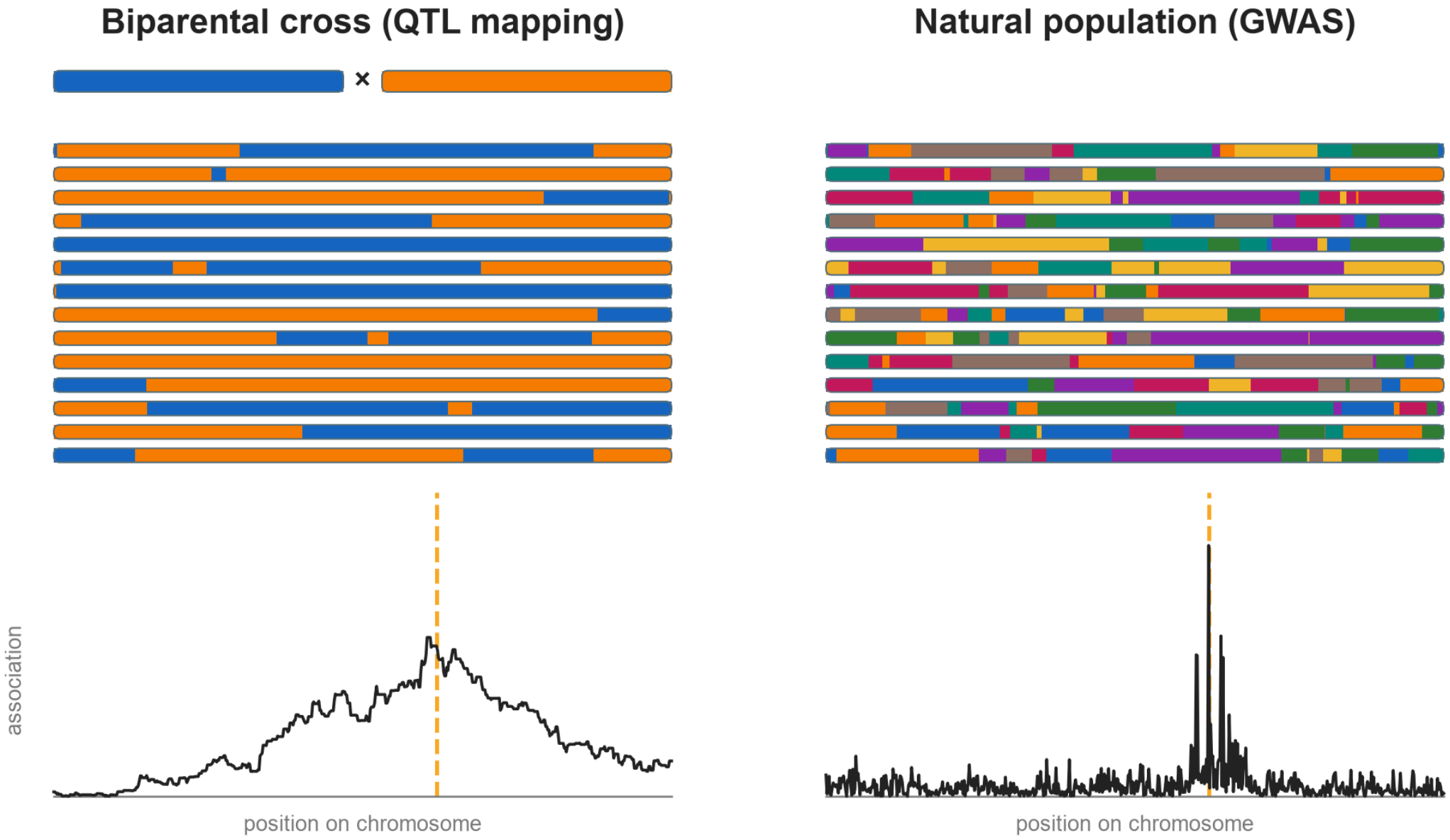
2 alleles · few crossovers · long blocks

## Natural population (GWAS)

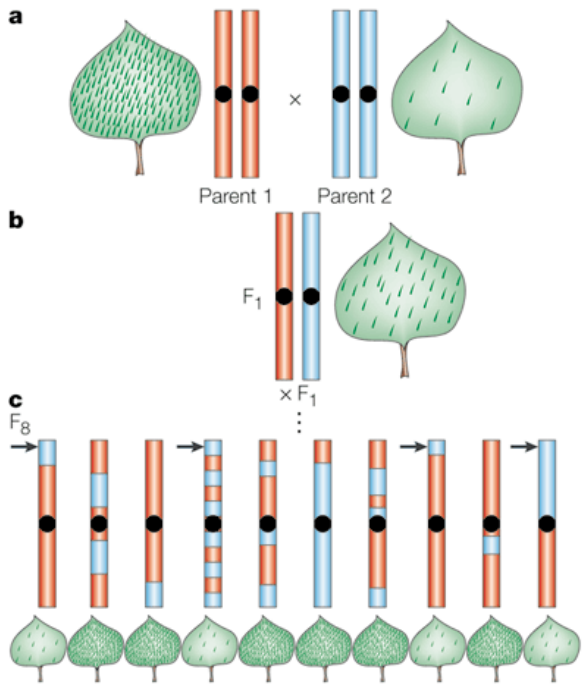


many alleles · many crossovers · short blocks

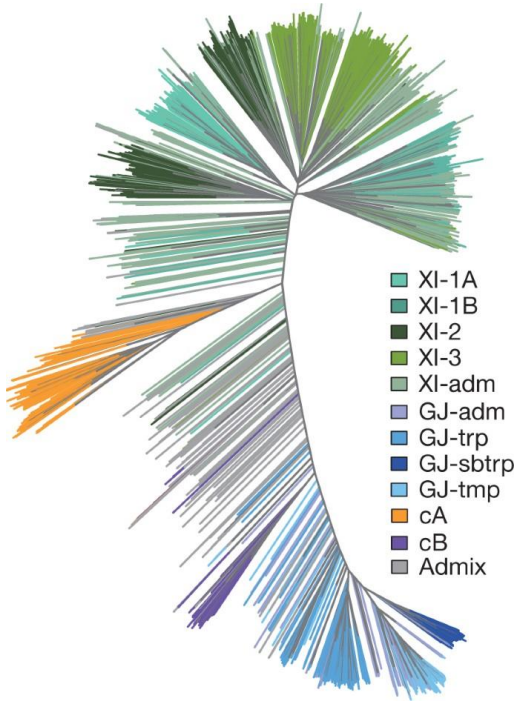
# Long blocks → wide peak



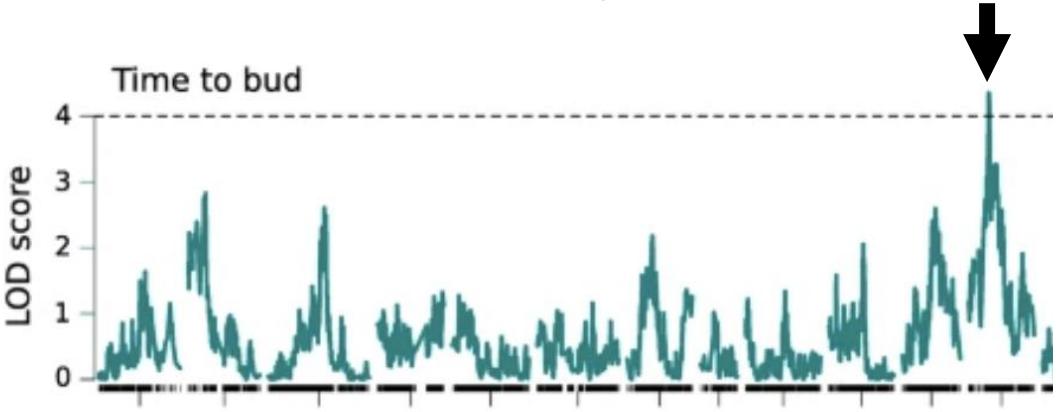
# QTL Mapping Population



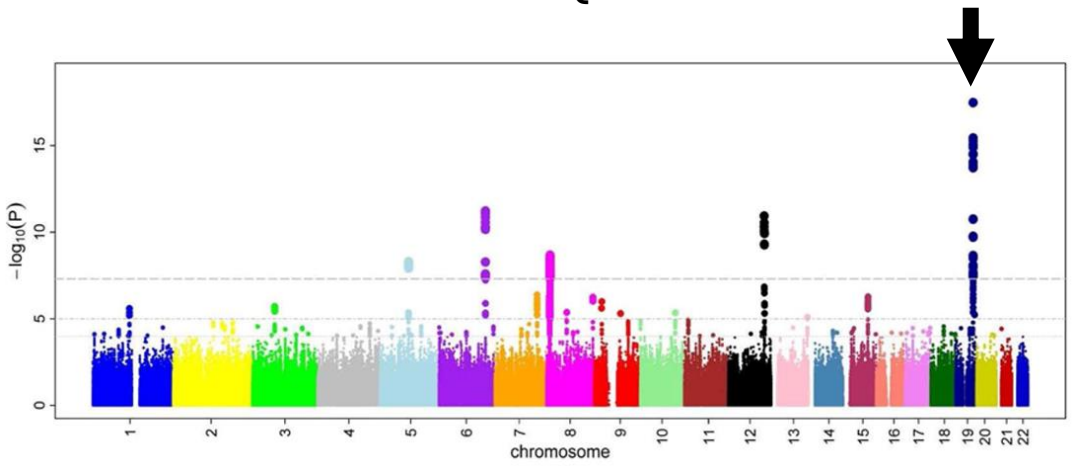

# GWAS Population




"Quantitative Trait Locus"




"Quantitative Trait Locus"

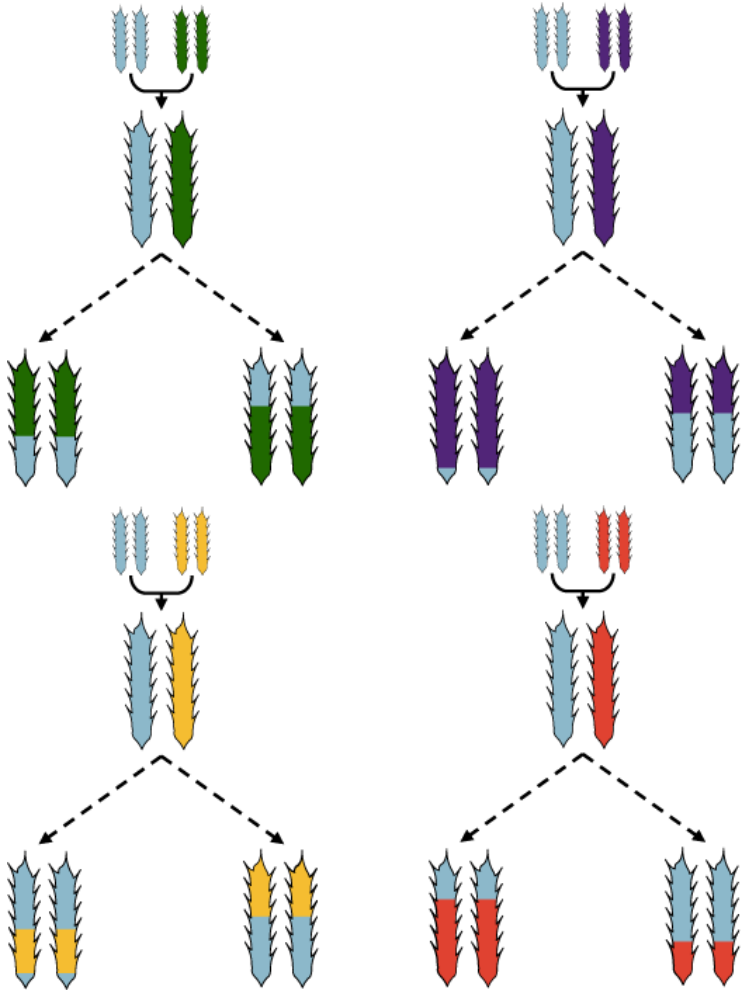


# QTL mapping vs GWAS

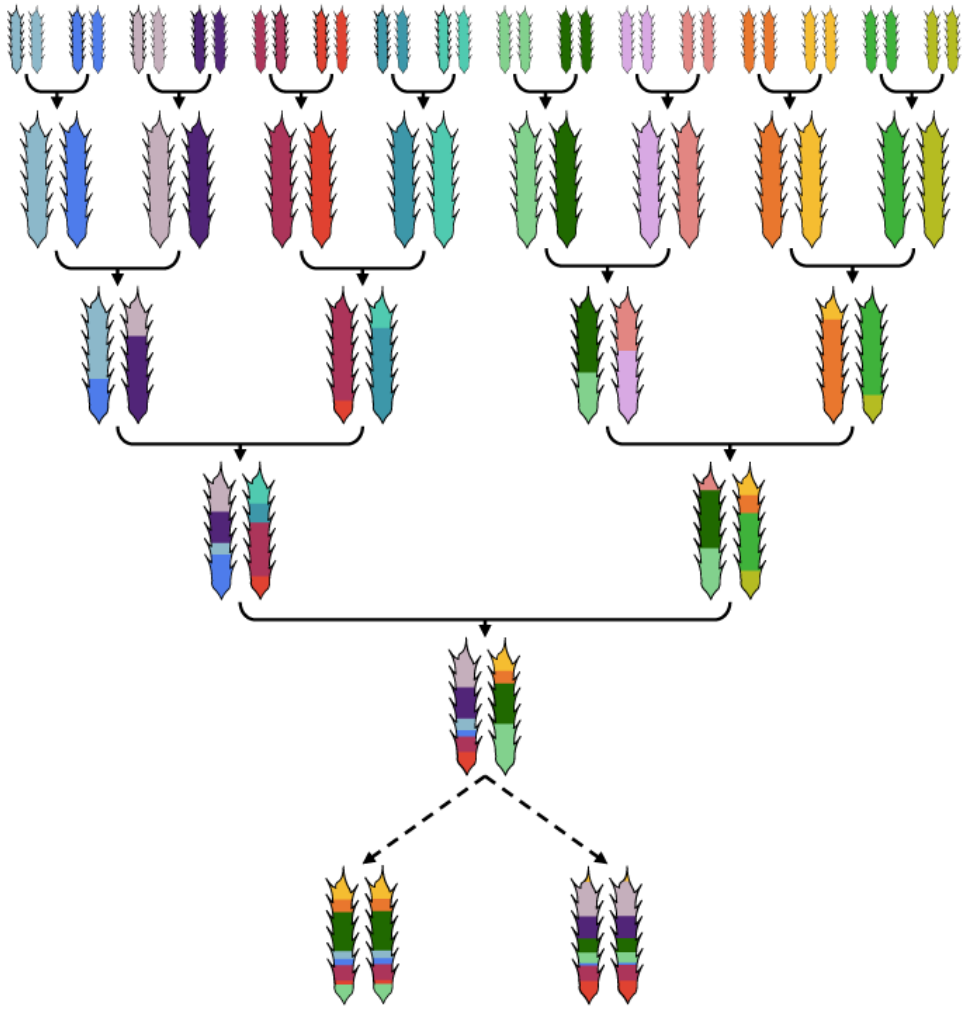
| | QTL mapping | GWAS |
|---|---|---|
| population | a cross you make (RILs) | lines nature made |
| alleles | 2 | many |
| peak | wide | narrow |
| pop. structure | none | false positives |
| the big win | clean design | genotype once,<br>phenotype many times |

# Multi-parent Mapping Populations: controlled relatedness of QTL mapping + allelic diversity of GWAS

Nested Association Mapping (NAM) panel



Multi-parent Advanced Generation Inter-Cross (MAGIC)



Is it “QTL mapping” or “GWAS”?



# Go to google scholar

Search “GWAS in _____”

*(your favorite plant/crop)*

Copy the title of a paper that sounds interesting, for future reading potentially

Post paper title to the live poll

# Live poll

Text wall

**Search Google Scholar for "GWAS in ___" (your favorite plant or crop). Paste the title of a paper that sounds interesting.**

Scan with your phone

Join on your phone or laptop: greymonroe.github.io/live-poll-test/submit.html?poll=pls152-fg-gwaspapers

host screen

# Take-home: phenotype → gene

| | |
|---|---|
| **Mutant screen** | break genes, find what they do |
| **Complementation** | same gene, or different genes? |
| **MutMap** | pool the mutants, find the region |
| **QTL mapping** | cross two lines: 2 alleles, wide peak |
| **GWAS** | nature's lines: many alleles, narrow peak |

**Every peak is a region, not a proof**